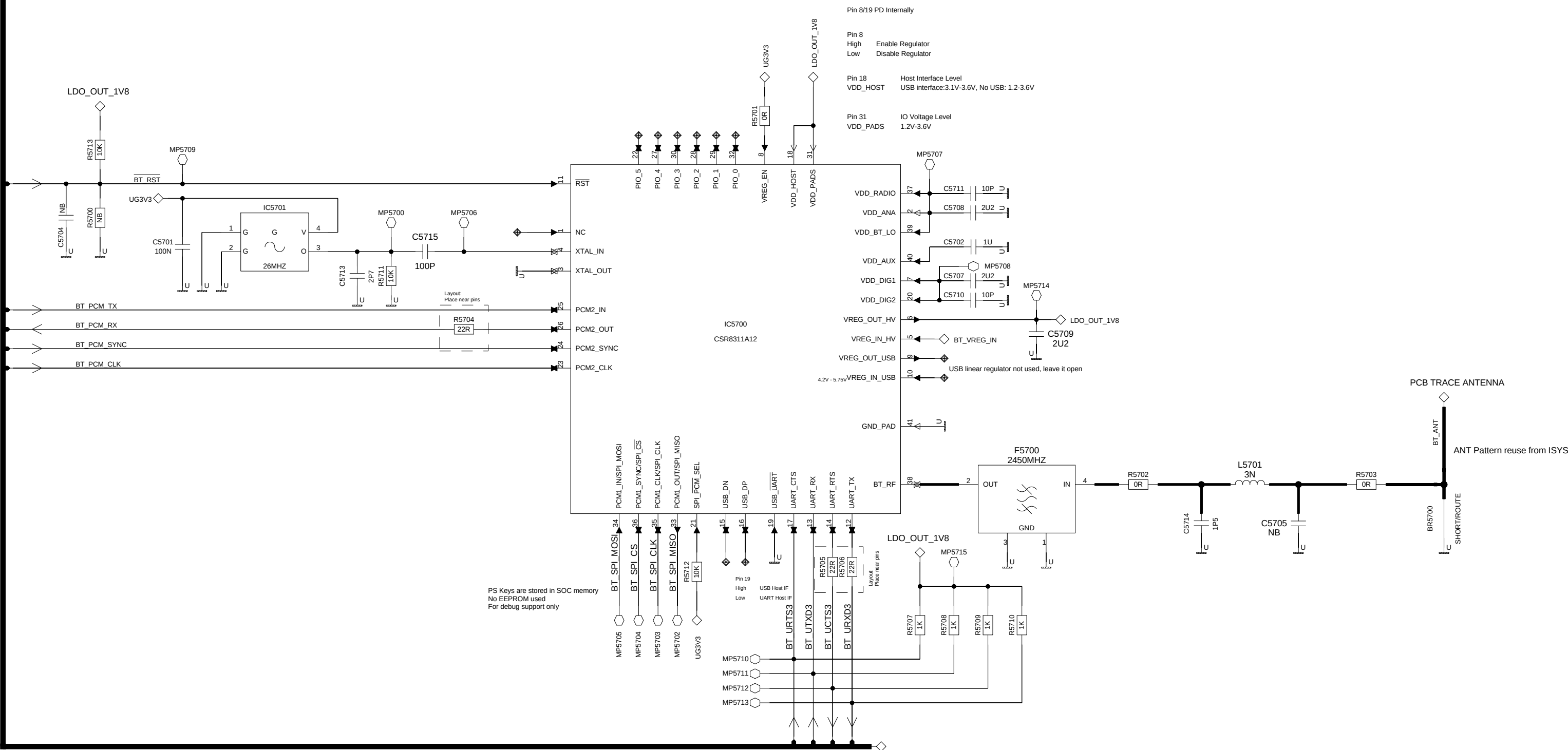
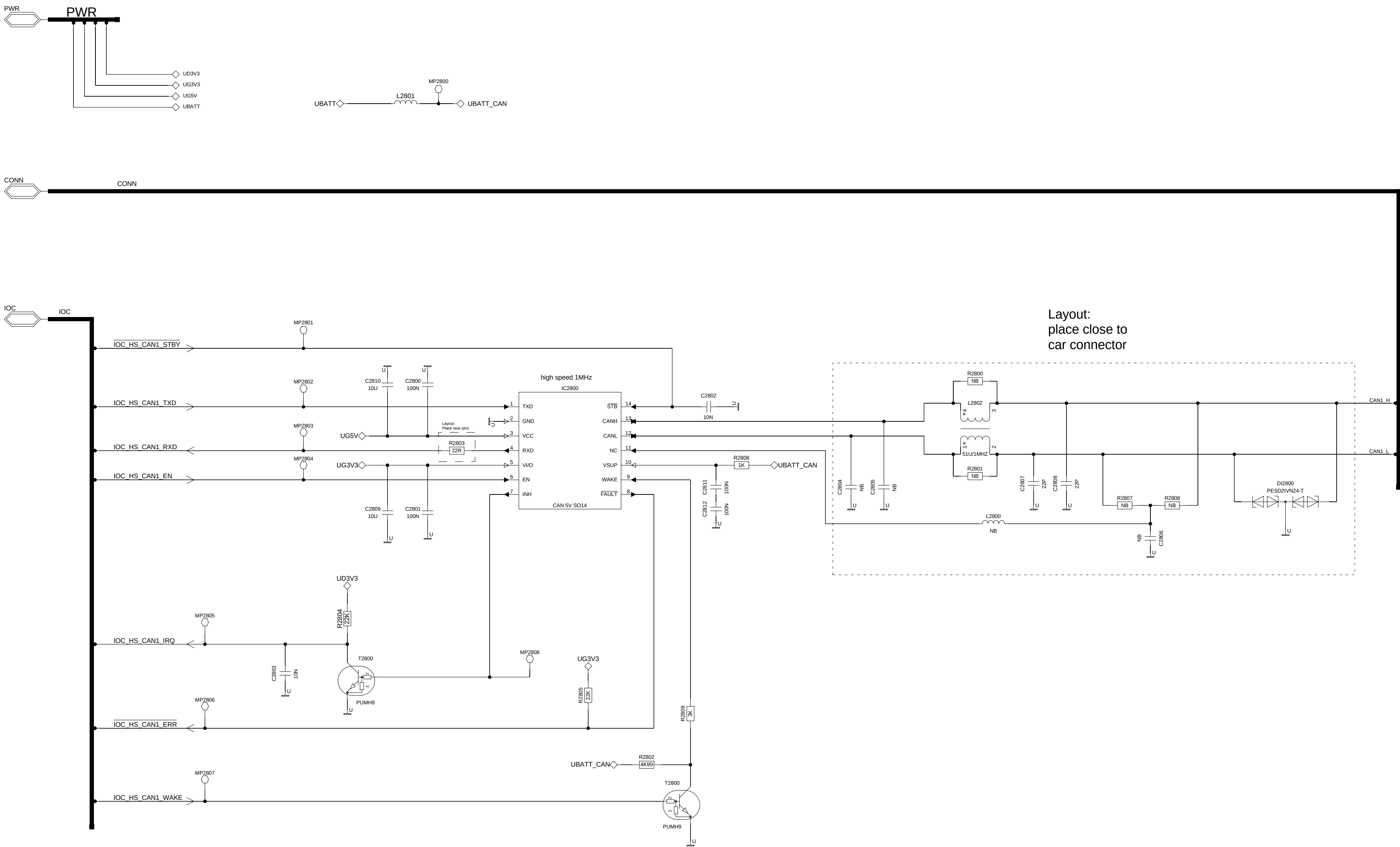


[illegible]

Rev.	type of change			change-no.	date of change	auth.	checked
		Copyright reserved	Wiring modifications reserved		date	name	
				created			
				checked			
schematic				Sheet	Sheets of	Sample	
4010_Bluetooth				2	31		
substitute for:				Type	Variant	Part Number	
substitute by:					C143		

CAN (2800-2899)



Layout:
place close to
car connector

Rev.	type of change		change-no.	date of change	auth. checked
		Copyright reserved	Wiring modifications reserved	date name	
				created	
				checked	
schematic			Sheet	Sheets of	Sample
4010_CAN			3	31	
substitute for:		substitute by:	Type	Variant	Part Number

The schematic diagram illustrates the ATMX1057T-A19 board, centered around the AT7700 microcontroller. The board includes several key components and connections:

- Microcontroller (AT7700):** The central component, with pins for power, ground, and various control signals.
- Power Management:** Includes a 3.3V regulator (UG3V3 L7000) and a 5V regulator (UG3V3 L7001). The AT7700 is powered by a 3.3V supply (VDD_3.3) and a 5V supply (VDD_5).
- Memory:** Includes a 16MB flash memory (MP7000) and a 16MB SRAM (MP7005).
- Control Signals:** Includes a reset signal (RST) and a touch signal (TOUCH).
- Connectors:** Includes a 10-pin header (TP_10P) and a 20-pin header (TP_20P).
- Other Components:** Includes a 10k pull-up resistor (R7000), a 10k pull-down resistor (R7001), and a 10k pull-up resistor (R7002).

The diagram shows the internal connections between these components and the AT7700 module, including the AT7700 module's internal components and the AT7700 module's external components.

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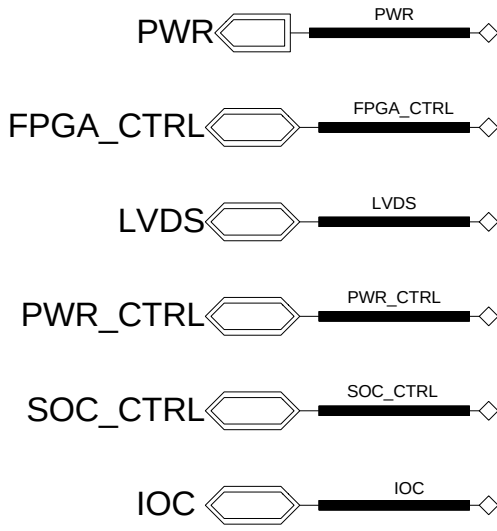
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				created	
				checked	
schematic			Sheet	Sheets of	Sample
4010_Debug			5	31	
substitute for:			Type	Variant	Part Number
substitute by:					

1) - Route as differential pair signals.
 - Differential impedance: 100ohm (±10%)
 - Maximum intra-pair skew: 0.127mm (5mils)

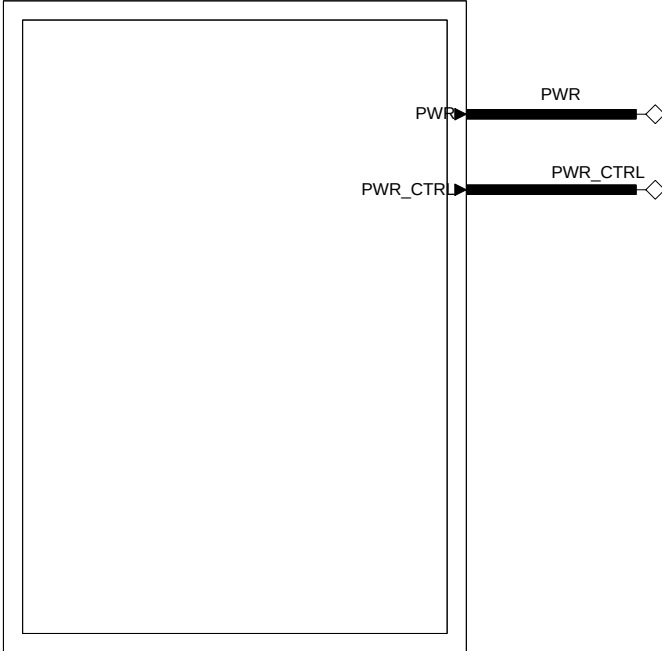
Marvell recommends using Varistors for ESD protection.
 TDK part # AVRH10C101KT1R1NE8

Marvell recommends using Varistors
for ESD protection.
TDK part # AVRH10C101KT1R1NE8

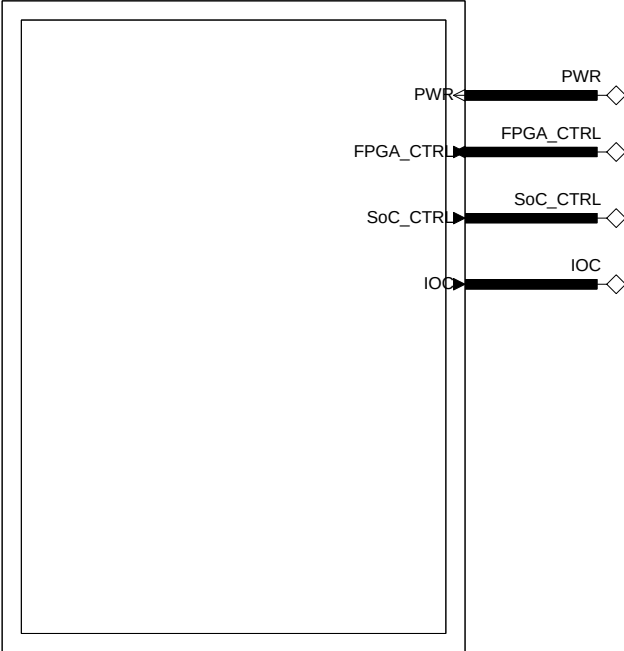
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schematic 4010_ETH_PHY			Sheet	Sheets of	Sample
			6	31	
substitute for:			Type	Variant	Part Number
substitute by:					



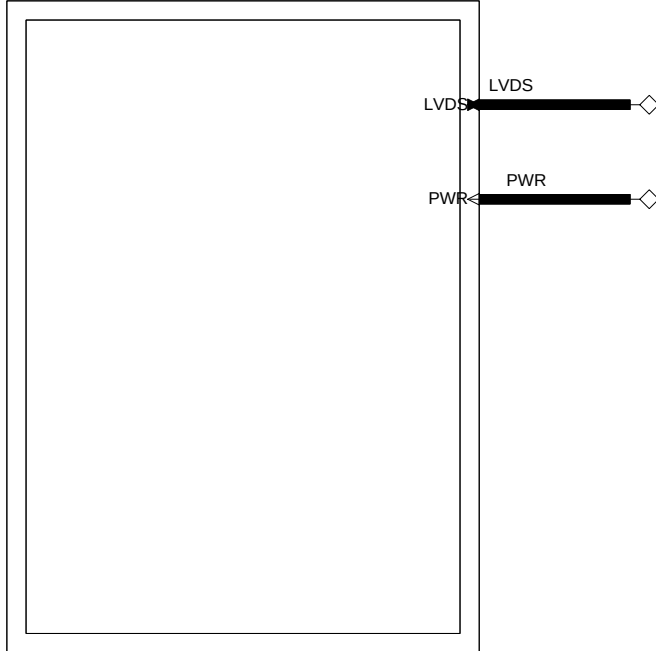
FPGA_POWER



FPGA_CONF

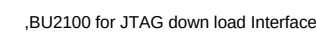
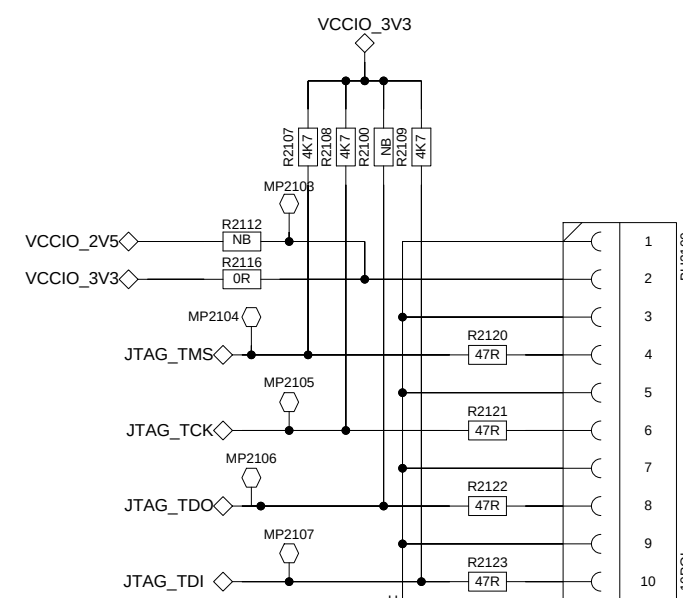
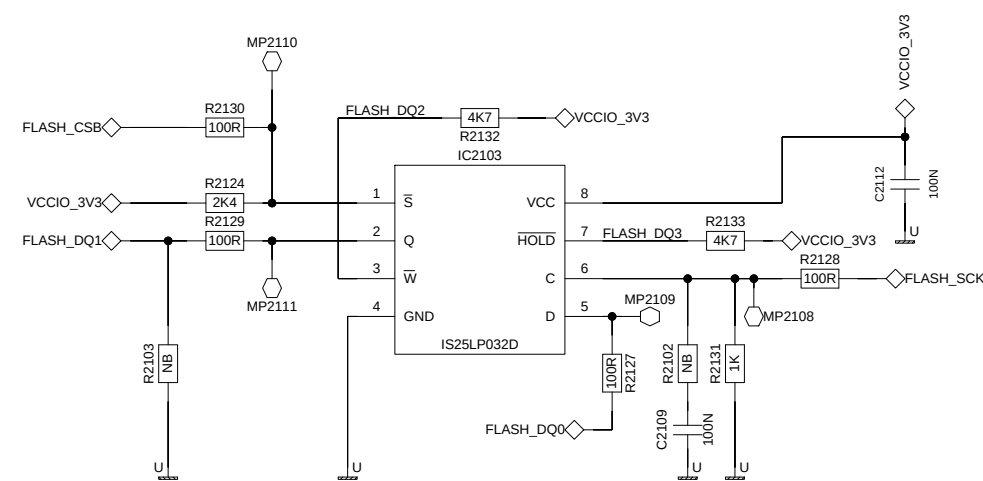
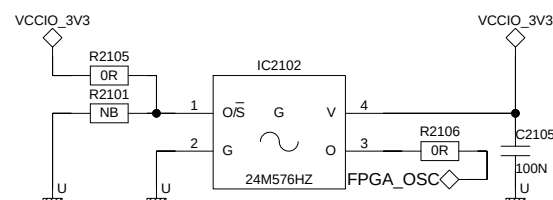
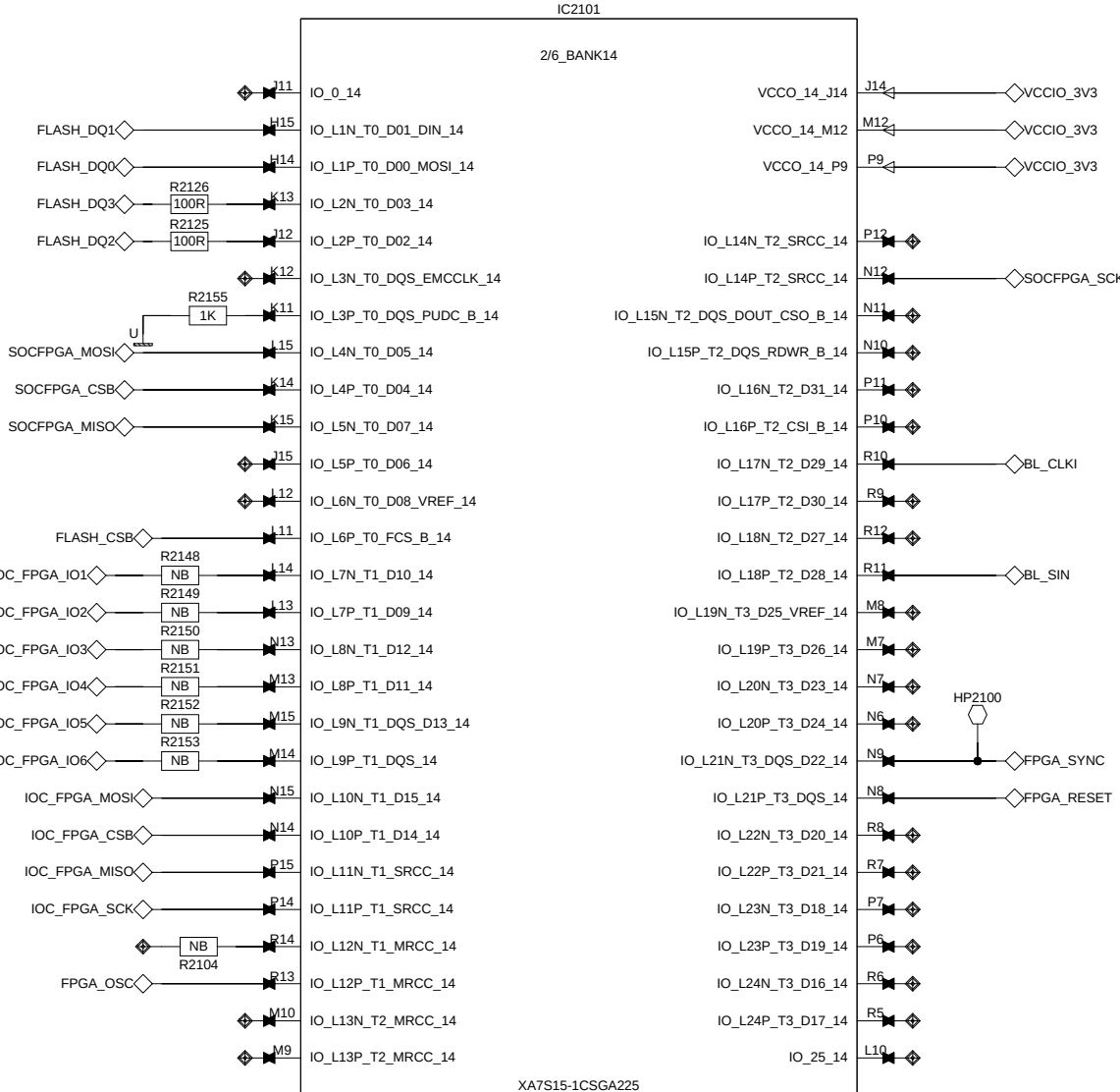
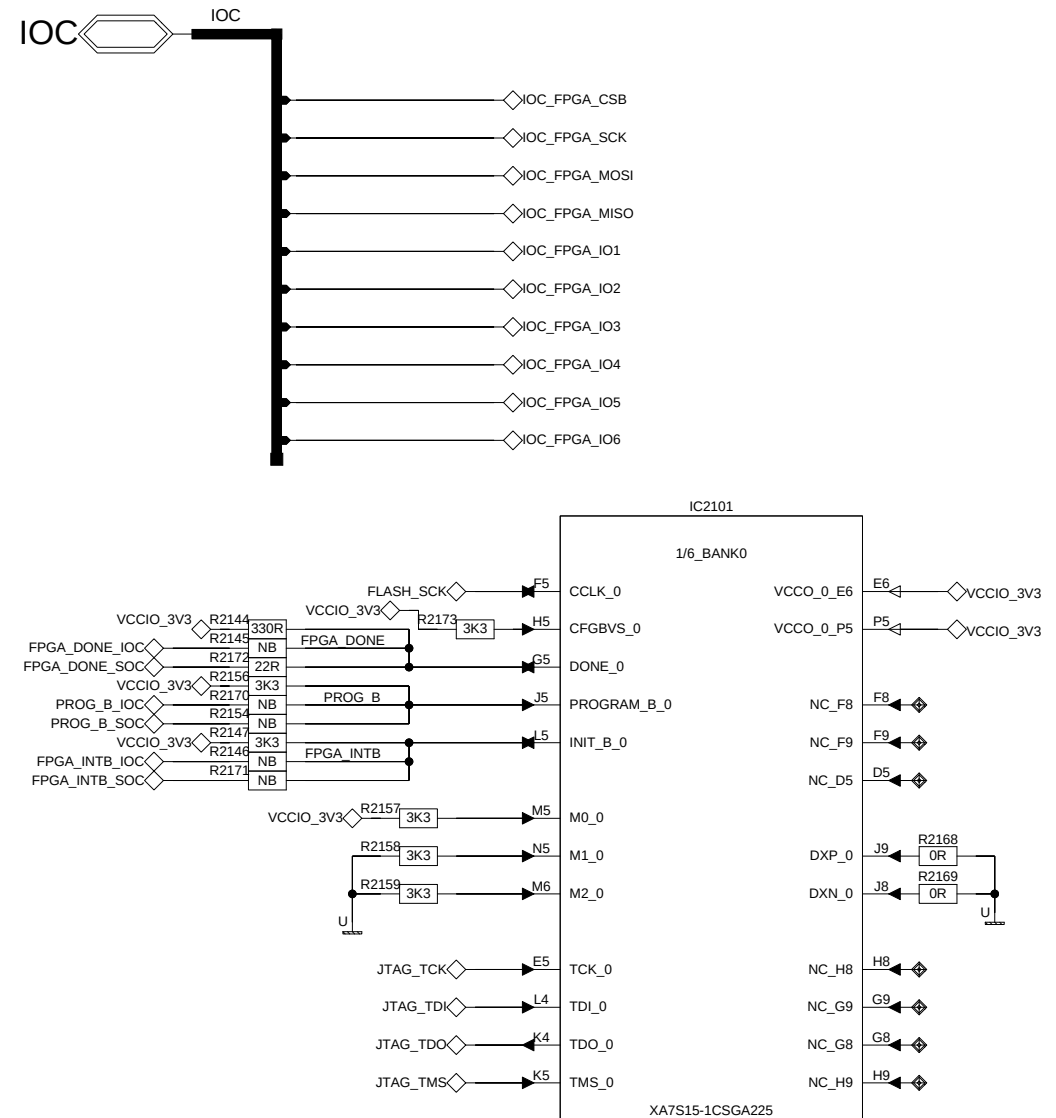
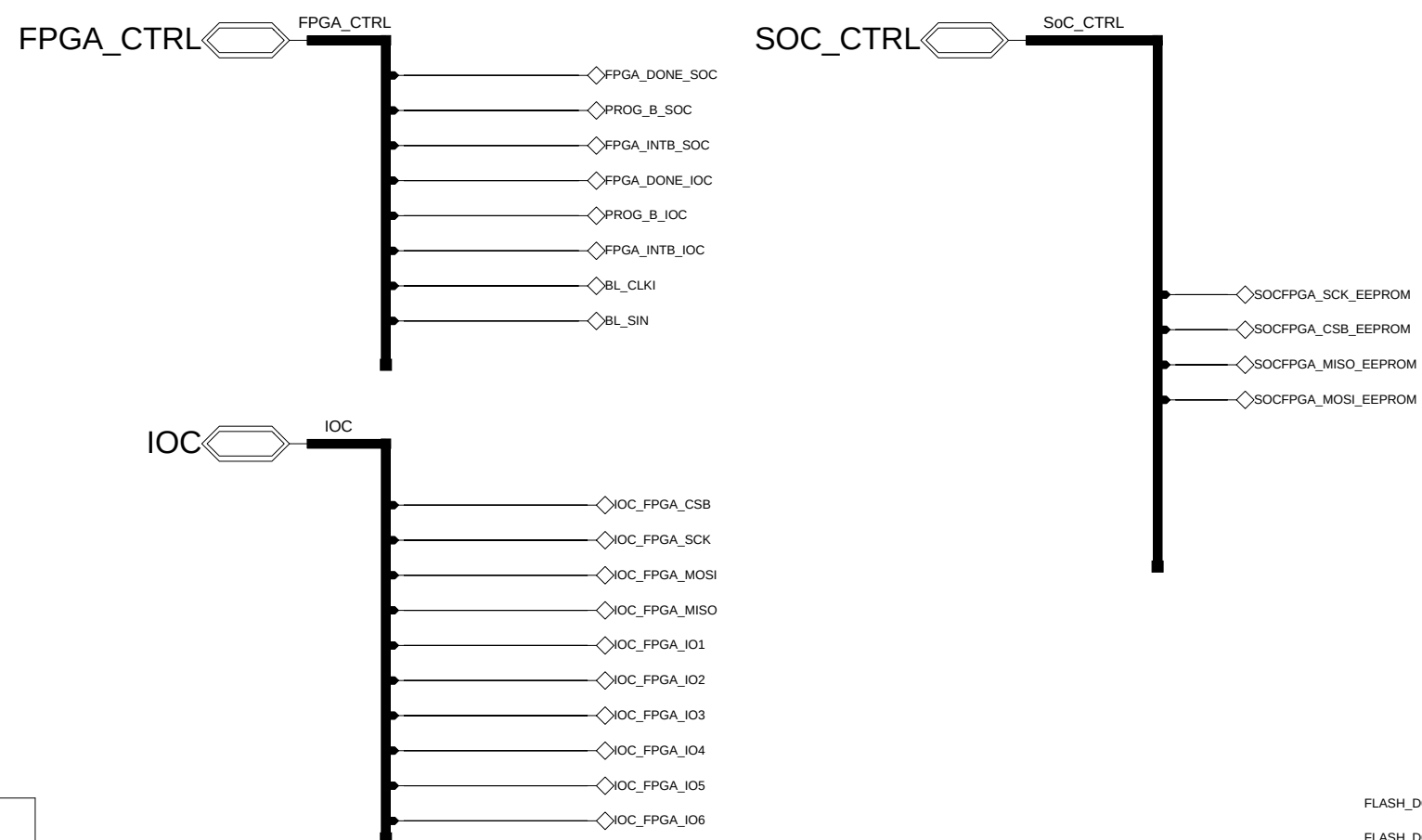


FPGA_VIDEO

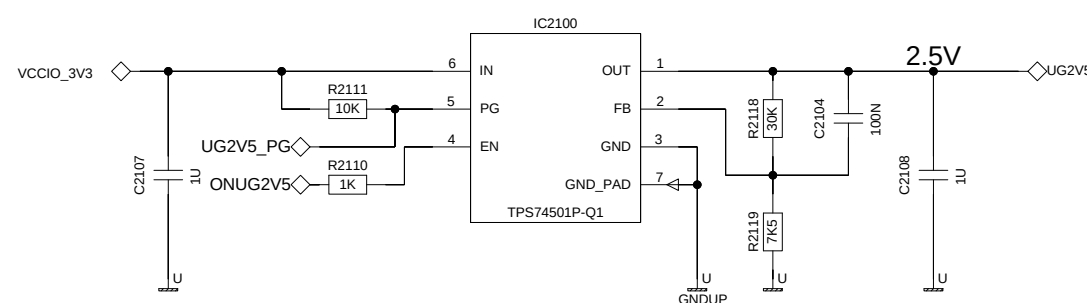
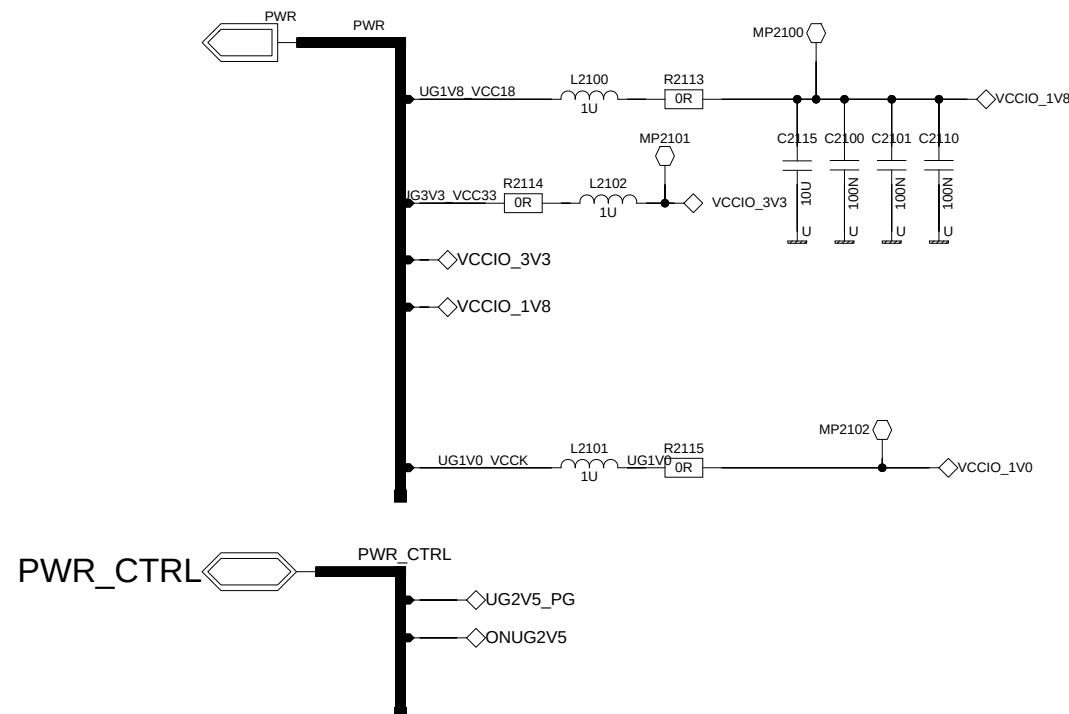
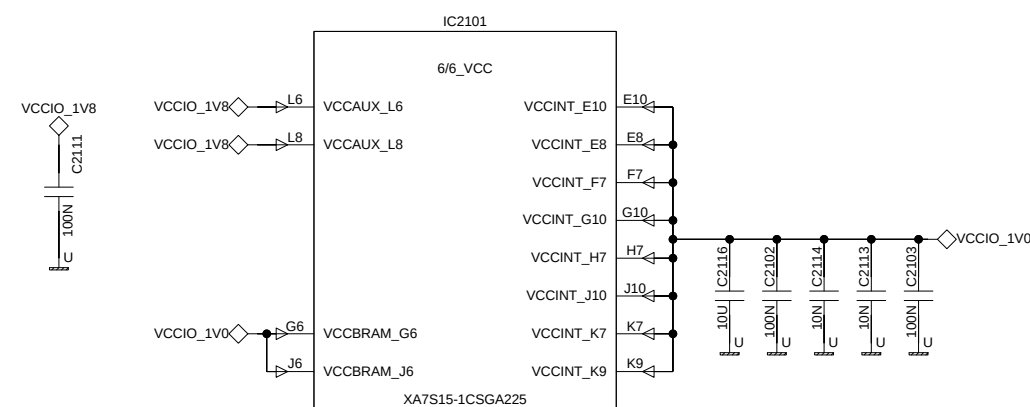
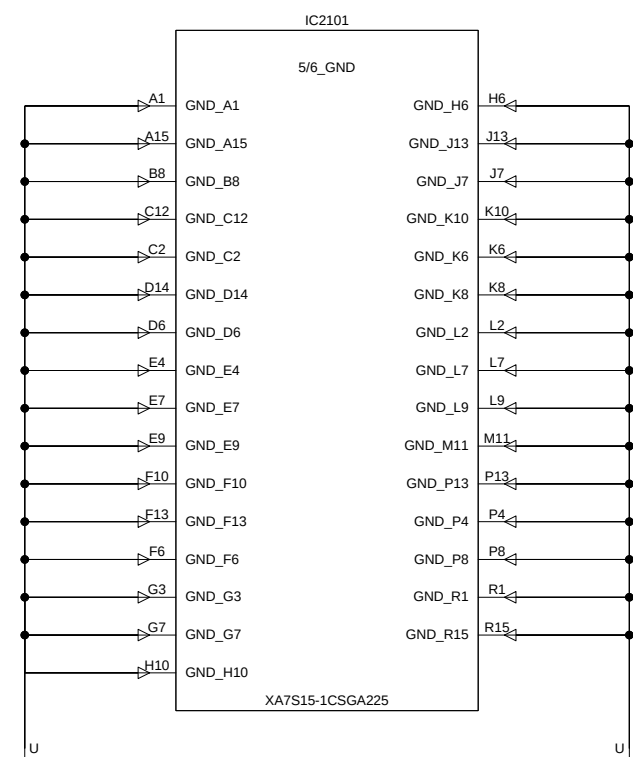


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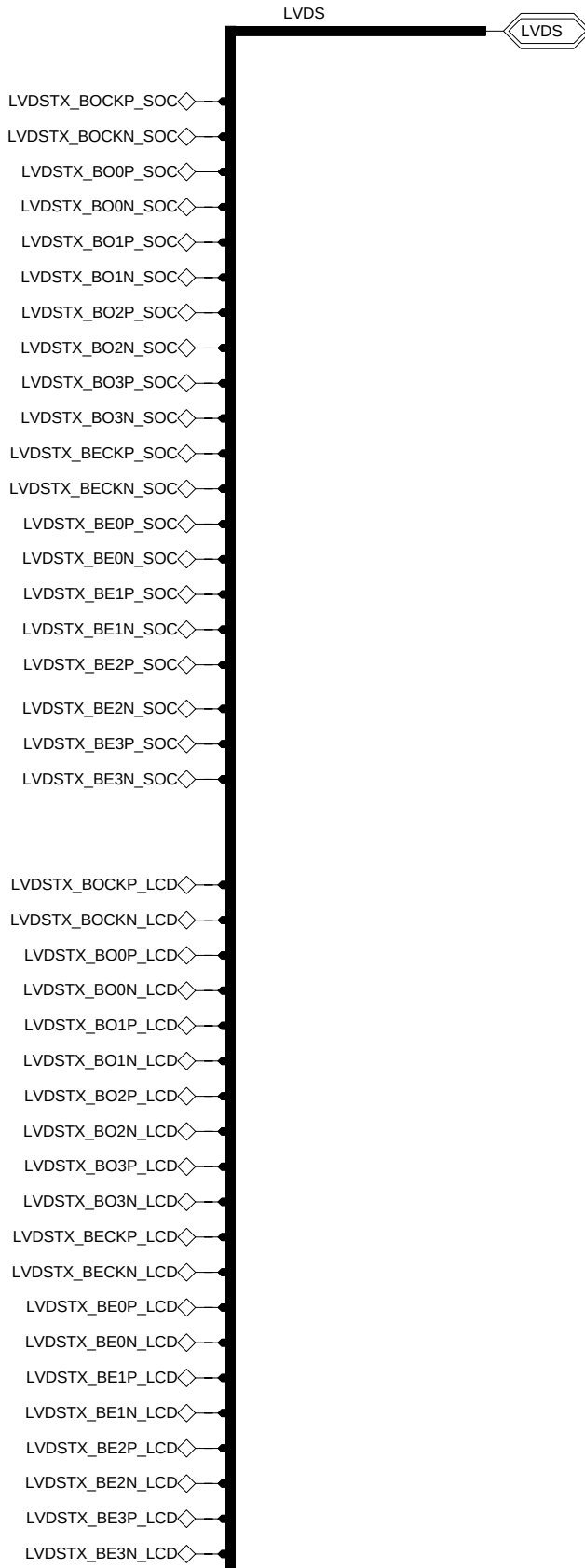
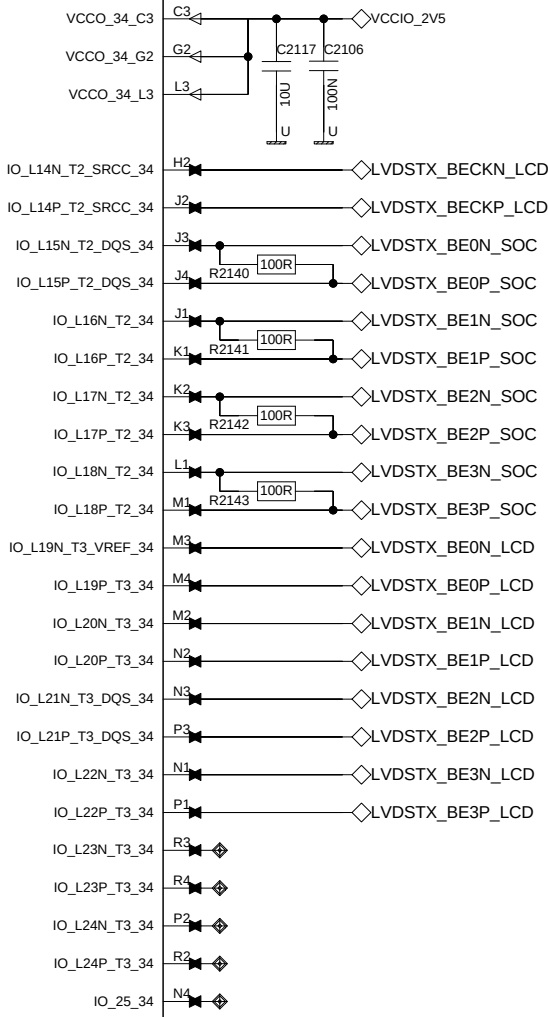
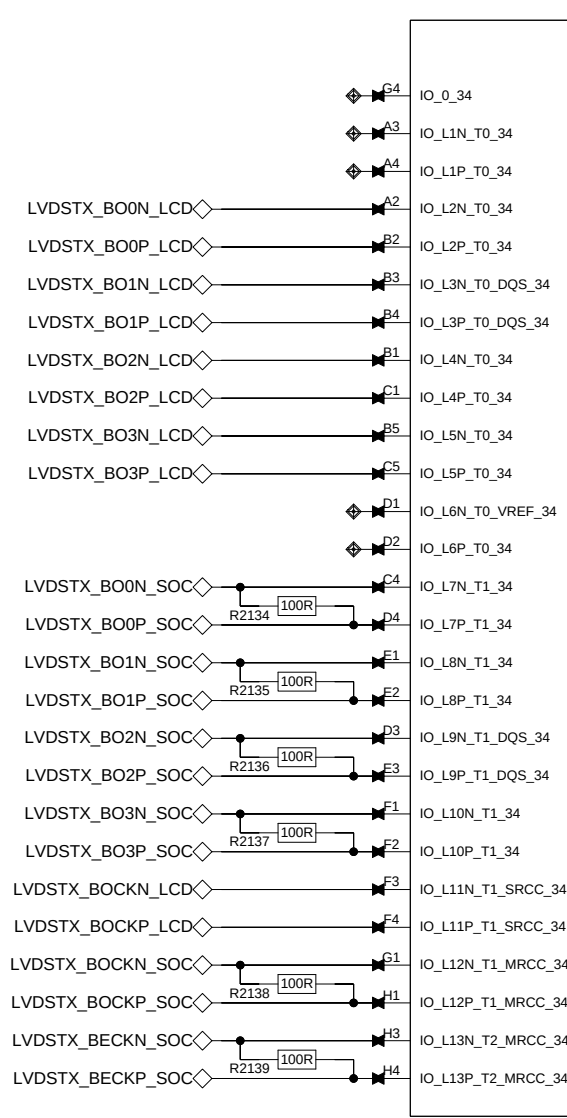
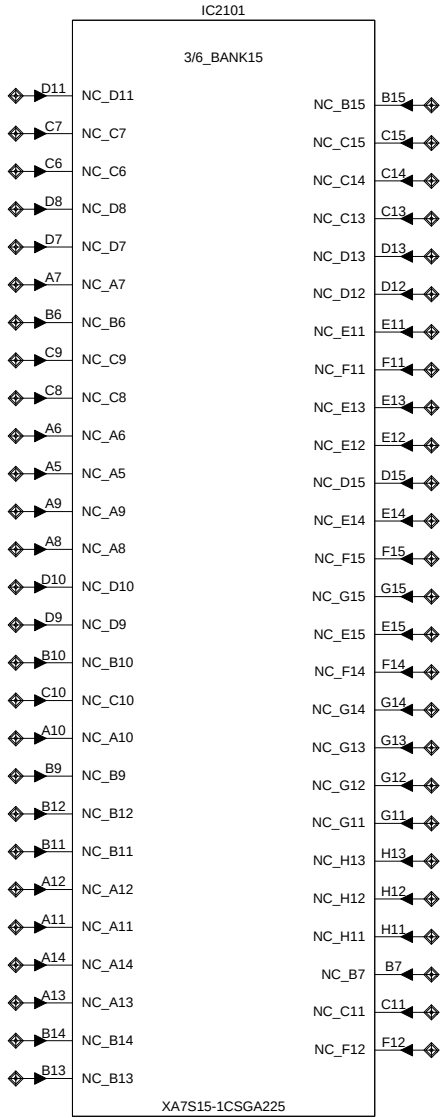
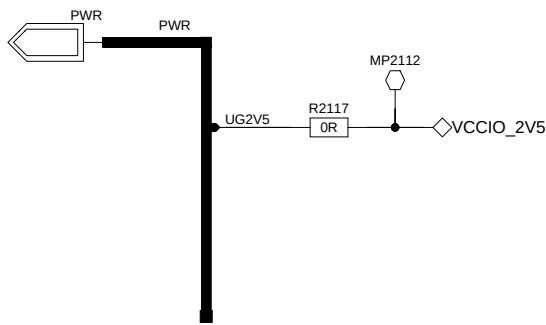
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schematic:					Sheet 7	Sheets of 31		Sample	
4030_FPGA					Type	Variant C143		Part Number	
substitute for:			substitute by:						



Rev.	type of change		change-no.	date of change	auth. checked
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FPGA_CONF			Type	Variant	Part Number
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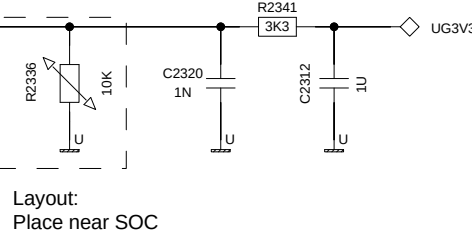


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					checked					
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FPGA_POWER						9	31			
substitute for:				substitute by:		Type	Variant		Part Number	
							C143			



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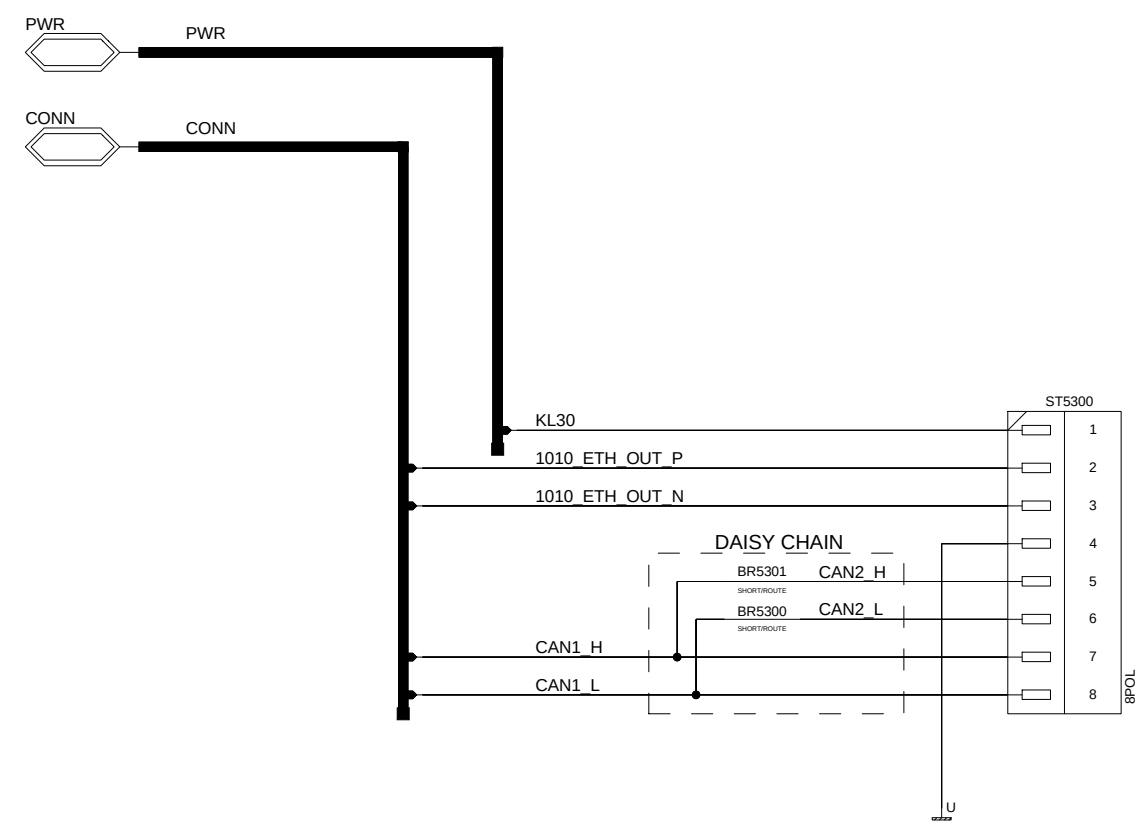
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				checked		
schematic			Sheet	Sheets of	Sample	
FPGA_VIDEO			10	31		
substitute for:			Type	Variant	Part Number	
				C143		



INTP0	AWO	IOC_BKLT_FAULT
INTP1	ISO	-----
INTP2	ISO	-----
INTP3	ISO	HS_CAN1_RXD
INTP4	ISO	-----
INTP5	AWO	-----
INTP6	AWO	-----
INTP7	AWO	-----
INTP8	AWO	HS_CAN1_IRQ
INTP9	AWO	-----
INTP10	ISO	IOC_SOC_IPC_IRQ
INTP11	AWO	-----
INTP12	AWO	-----
INTP13	AWO	-----
INTP14	AWO	DEBUG_RX
INTP15	AWO	-----

Rev.	type of change					change no.	date of change		auth.	checked
 Copyright reserved						date		name		
						created				
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schematic:						Sheet 11	Sheets of 31	Sample		
4010_IDC						Type	Variant C143		Part Number	
substitute for:						substitute by:				

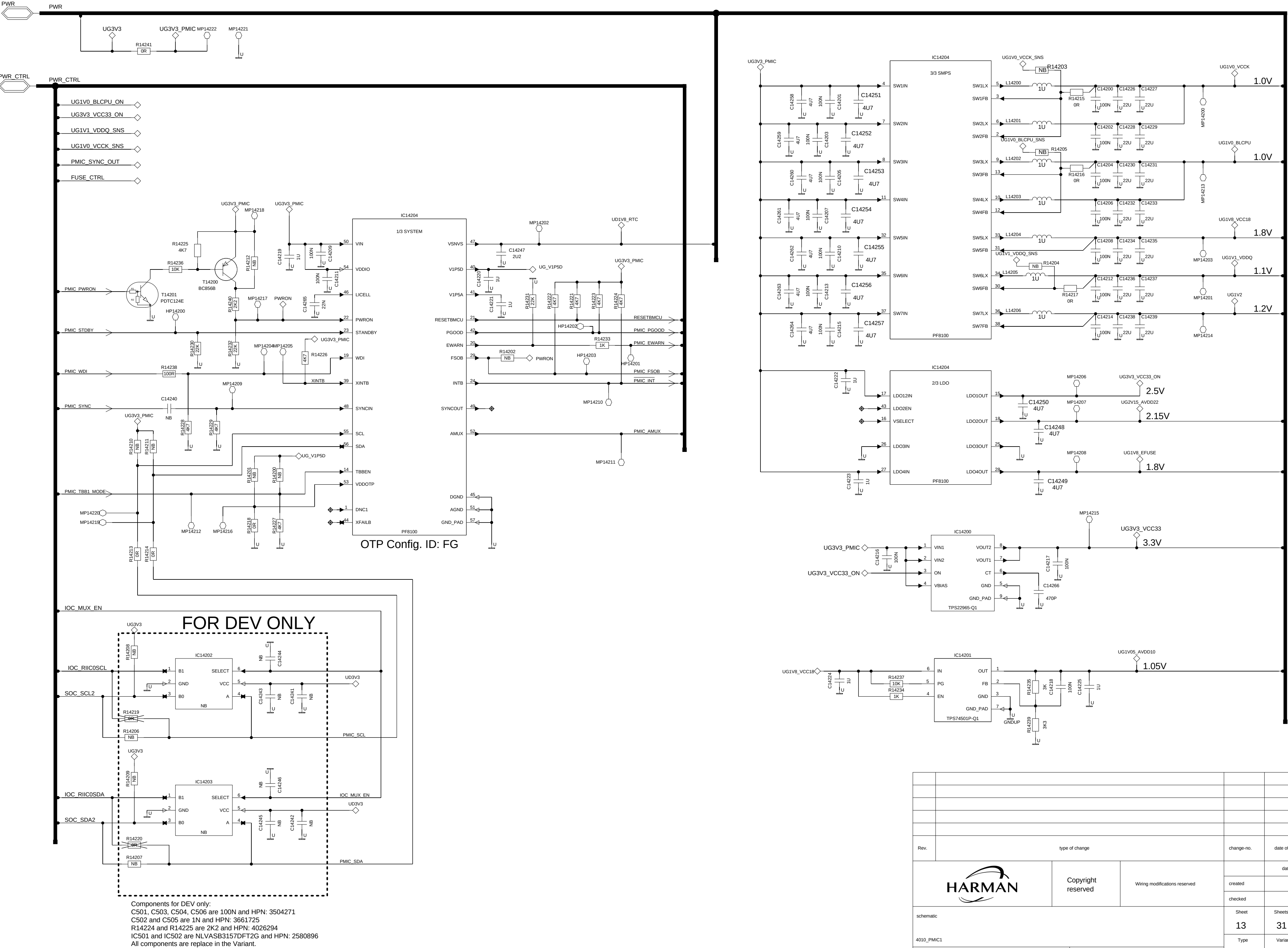
MAIN CONNECTOR (5300-5399)



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schematic			Sheet	Sheets of		Sample
4010_Main_Connector			12	31		
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substitute by:						

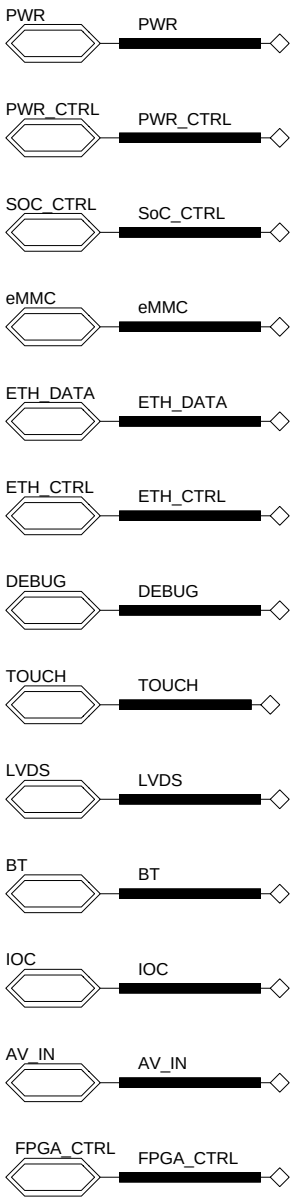
PMIC1 (14200-14299)



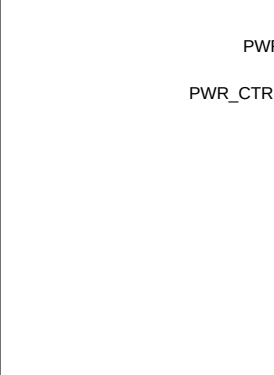
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substitute for:		substitute by:	Type	Variant	Part Number

SOC (10000-10999)

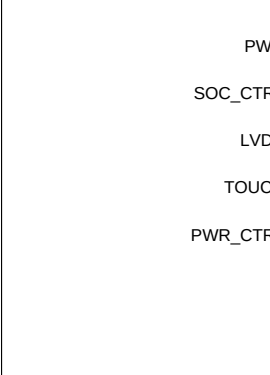


4010 SoC Power



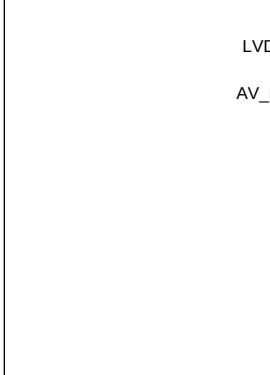
10500 - 10599

4010 SoC_CTRL



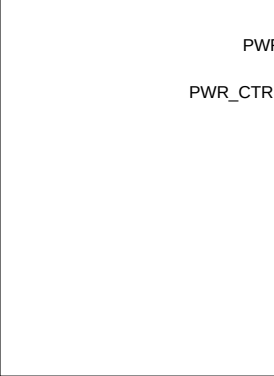
10600 - 10699

4010 SoC Video



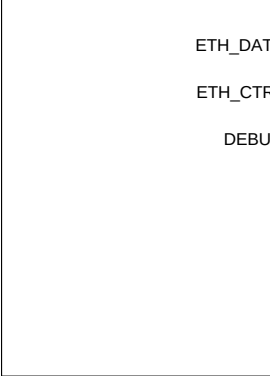
10700 - 10799

4010 SoC LPDDR4



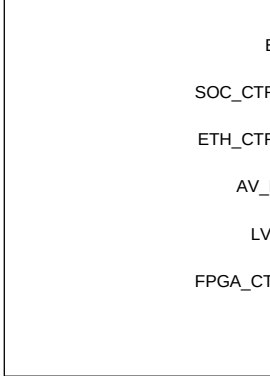
10400 - 10499

4010 SoC_ETH_PCIE_USB



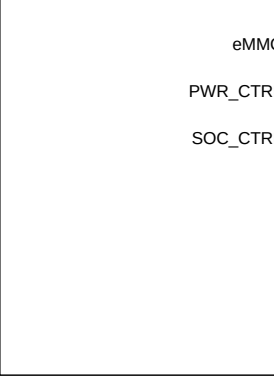
10300 - 10399

4010 SoC Audio



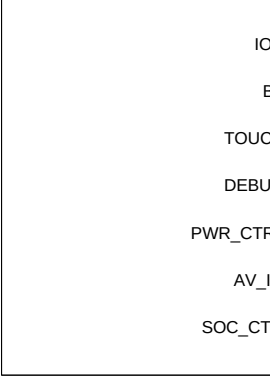
10800 - 10899

4010 SoC_eMMC_NAND_SD



10100 - 10199

4010 SoC_Serial_IF

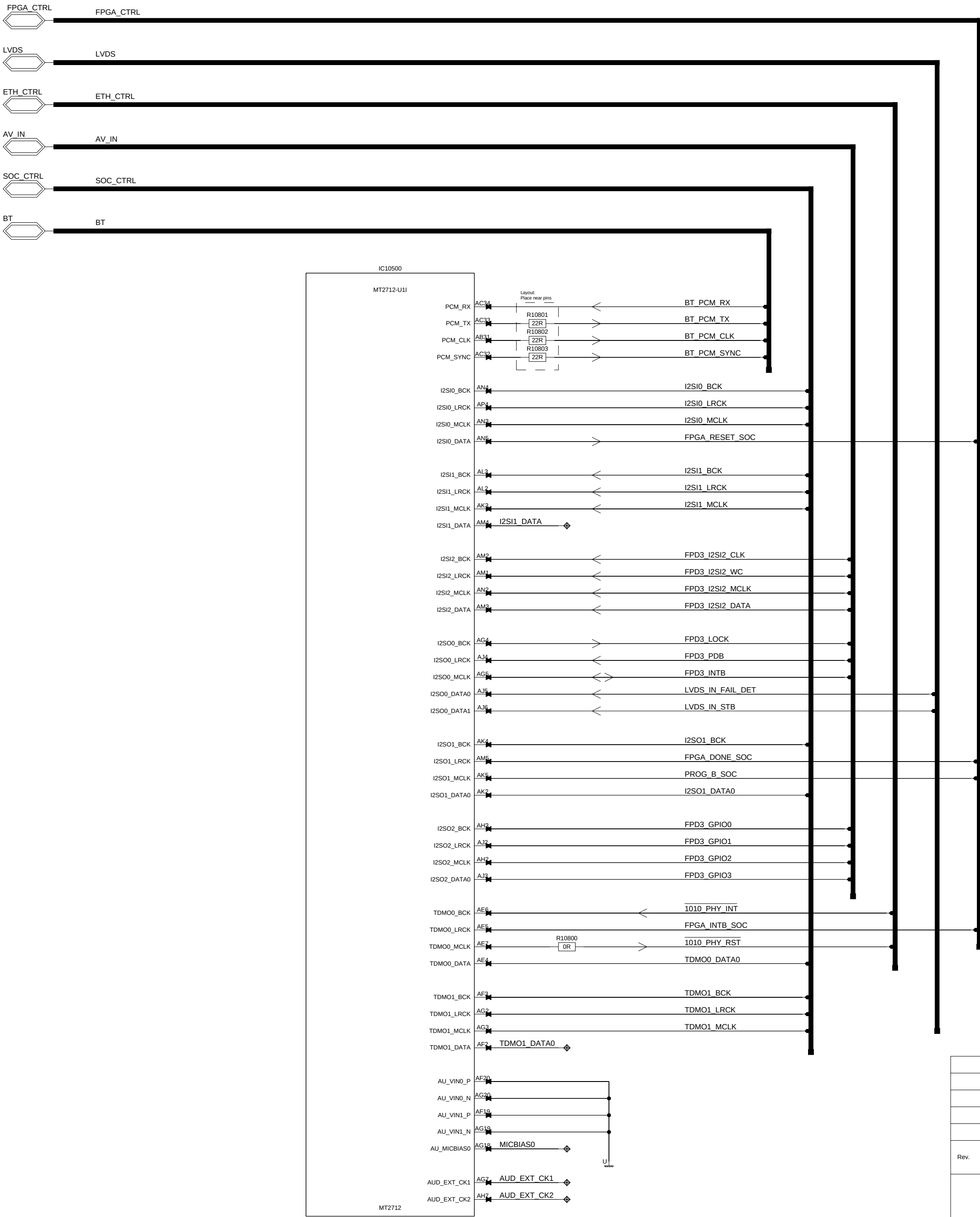


10200 - 10299

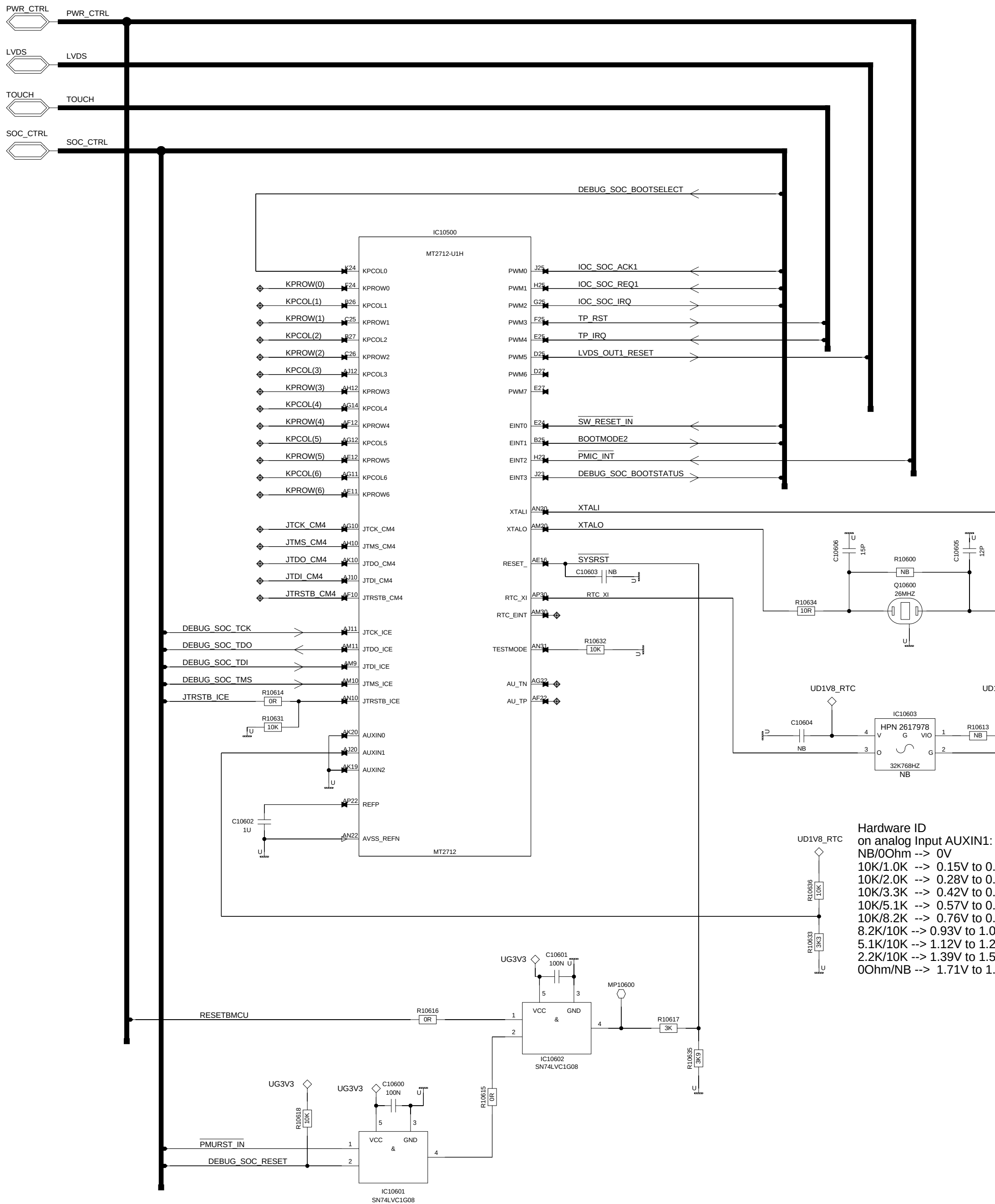
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				created		
				checked		
schematic			Sheet	Sheets of	Sample	
4010_SoC			15	31		
substitute for:			Type	Variant	Part Number	
			substitute by:			

SOC_AUDIO (10800-10899)



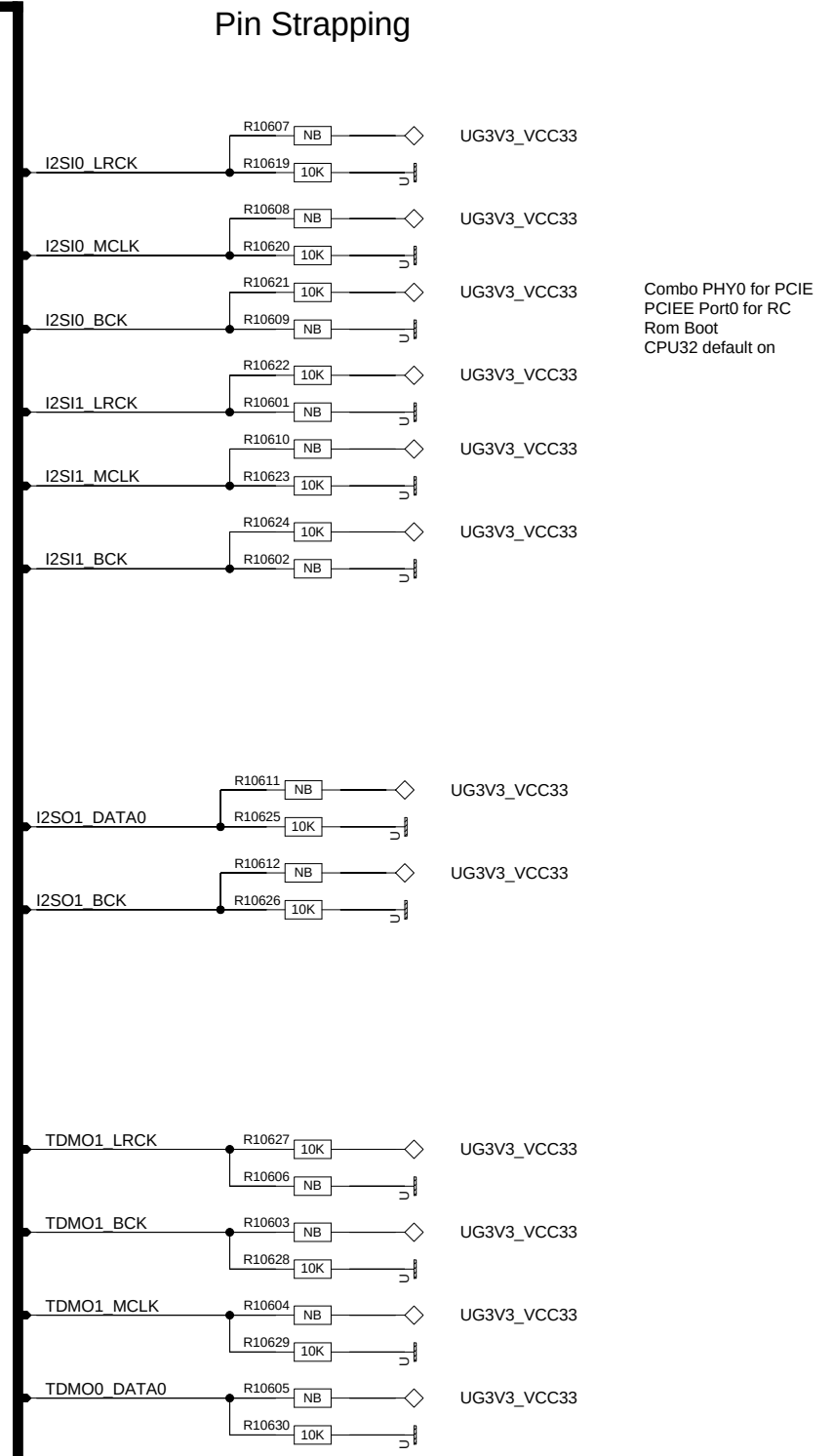
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schematic			Sheet	Sheets of	
4010_SoC_Audio			16	31	
substitute for:				Type	Variant
				Part Number	



	multi-used				multi-used	
Mode	I2Si0_LRCK	I2Si0_MCLK	I2Si0_BCK	I2Si1_BCK	I2Si1_LRCLK	I2Si1_MCLK
Combo PHY0 for USB	X	X	X	X	X	1
Combo PHY0 for PCIE	X	X	X	X	X	0
Combo PHY1 for USB	X	X	X	X	1	X
Combo PHY1 for PCIE	X	X	X	X	0	X
PCIEE Port0 for RC	X	X	X	1	X	X
PCIEE Port0 for EP	X	X	X	0	X	X
PCIEE Port1 for RC	X	X	1	X	X	X
PCIEE Port1 for EP	X	X	0	X	X	X
Rom Boot	X	0	X	X	X	X
Nor-flash bypass rom boot	X	1	X	X	X	X
CPU32 default on	0	X	X	X	X	X
CPU64 default on	1	1	X	X	X	X

i2S01_DATA0 status[1]	L	L	H	H
i2S01_BCK status[0]	L	H	L	H
1	EMMC	NAND	NOR	SD
2	SD	EMMC	NAND	NOR
3	NOR	SD	EMMC	NAND
4	NAND	NOR	SD	EMMC

Trapping Mode	TDMO1_LRCK	TDMO1_BCK	TDMO1_MCLK	TDMO1_DATA0
Slowest Boot device speed configuration	1	X	X	X
Fastest Boot device speed configuration	0	X	X	X
NAND randomizer: disable	X	0	X	X
NAND randomizer: enable	X	1	X	X
NOR flash address mode: 3-byte	X	X	0	X
NOR flash address mode: 4-byte	X	X	1	X
Alternative boot loader: disable	X	X	X	0
Alternative boot loader: enable	X	X	X	1

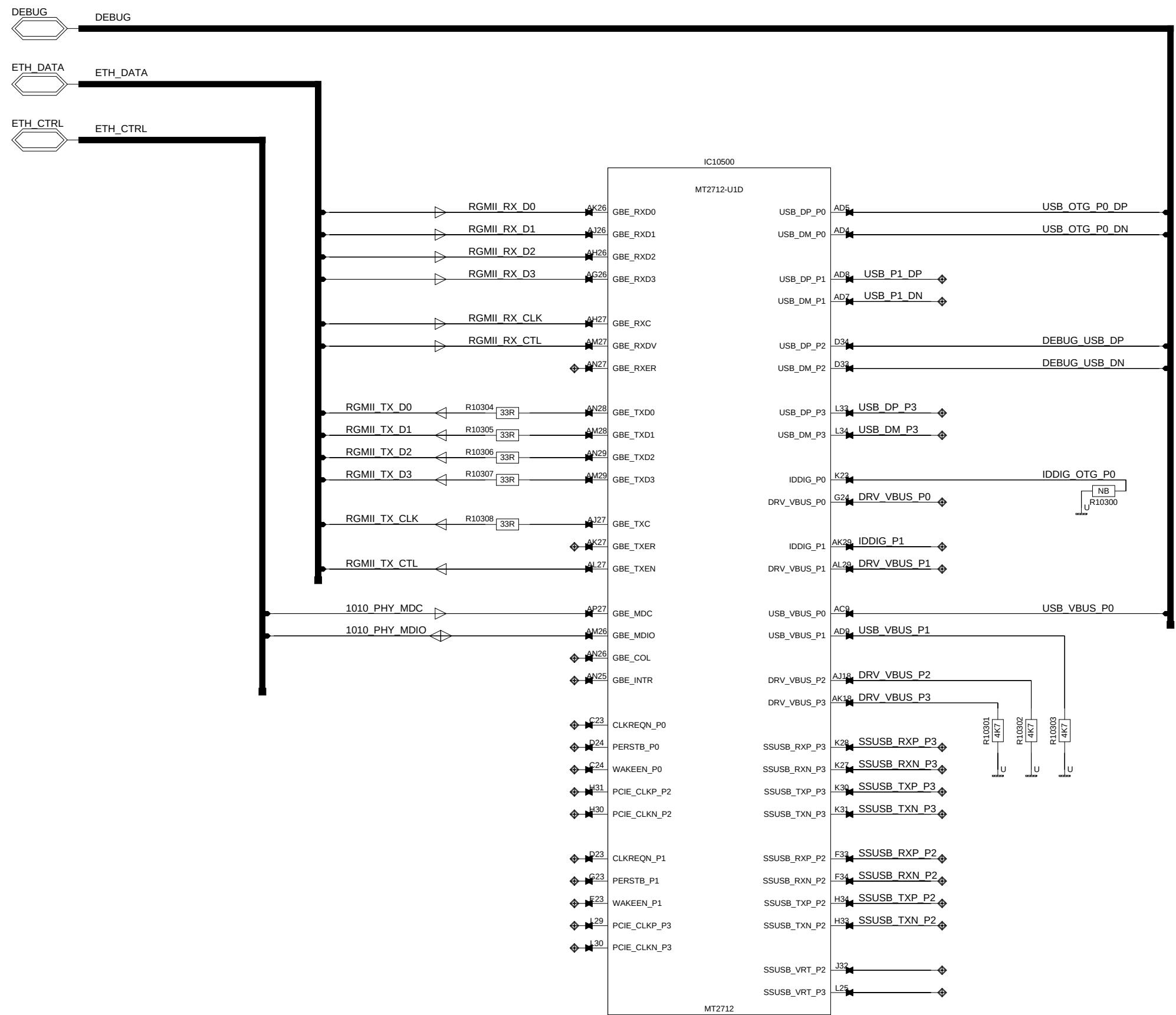


```
Hardware ID
NB/OOHm Input AUXIN1:
10K/1.0K --> 0.15V to 0.18V
10K/2.0K --> 0.28V to 0.32V
10K/3.3K --> 0.42V to 0.47V
10K/5.1K --> 0.57V to 0.65V
10K/8.2K --> 0.76V to 0.86V
8.2K/10K --> 0.93V to 1.05V
5.1K/10K --> 1.12V to 1.26V
2.2K/10K --> 1.39V to 1.55V
OOHm/NB --> 1.71V to 1.89V
```

ED1 HW
ED2 HW
ED3 HW
DV HW
PV HW
PV2 HW
PV3 HW

Rev.	type of change		change-no.	date of change	auth. checked
		Copyright reserved	Wiring modifications reserved	date name	
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schematic			Sheet 17	Sheets of	Sample
4010_SoC_CTRL			Type	Variant	Part Number
substitute for:		substitute by:			

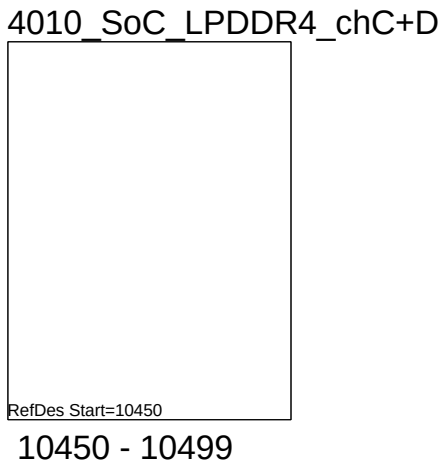
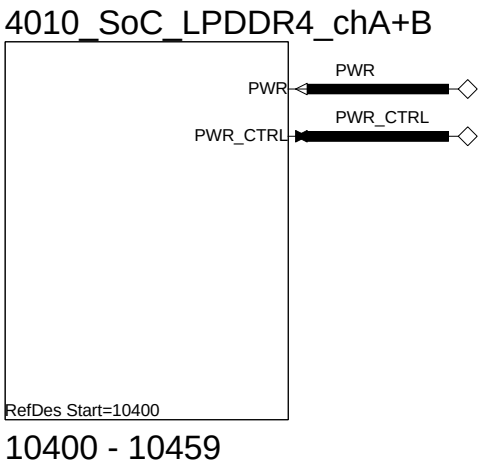
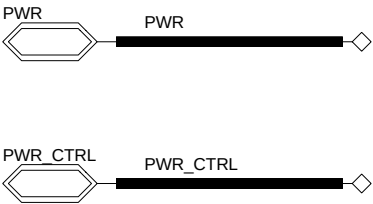
SOC_ETH_PCl_e_USB (10300-10399)



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schematic	Sheet		Sheets of		Sample
4010_SoC_ETH_PCl_e_USB	18		31		
substitute for:			Type	Variant	Part Number
substitute by:					

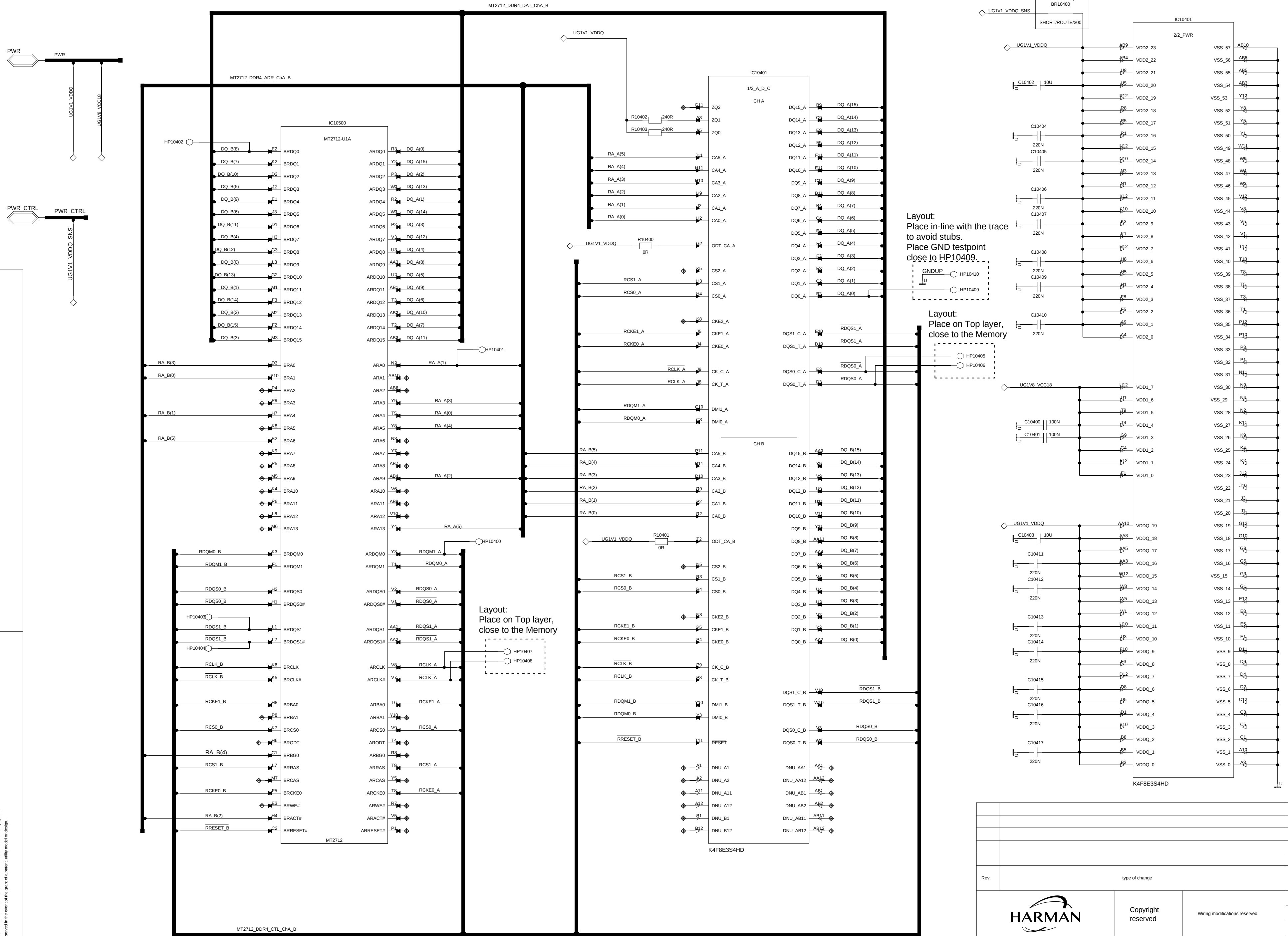
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SOC_LPDDR4 (10400-10499)



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		Copyright reserved	Wiring modifications reserved	date		name
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schematic			Sheet	Sheets of		Sample
4010_SoC_LPDDR4			19	31		
substitute for:			Type	Variant		Part Number

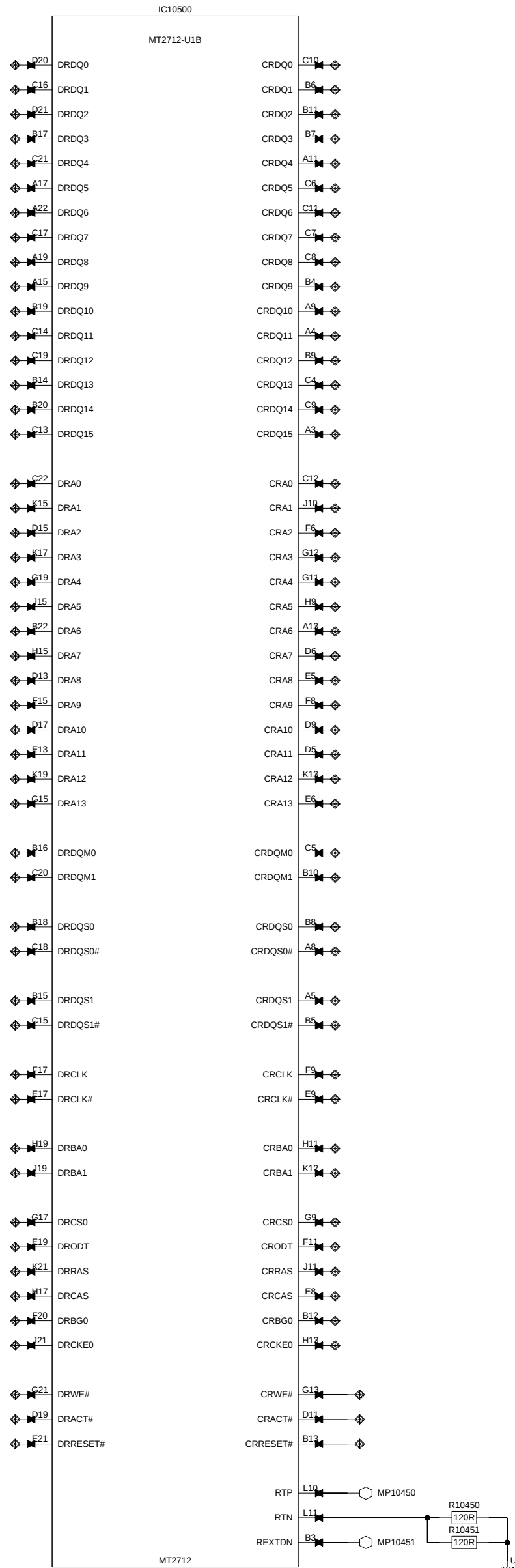
SOC_LPDDR4 (10400-10449)



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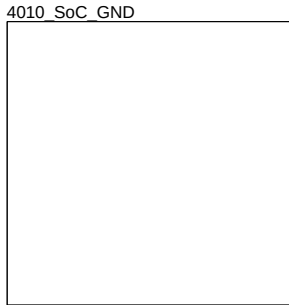
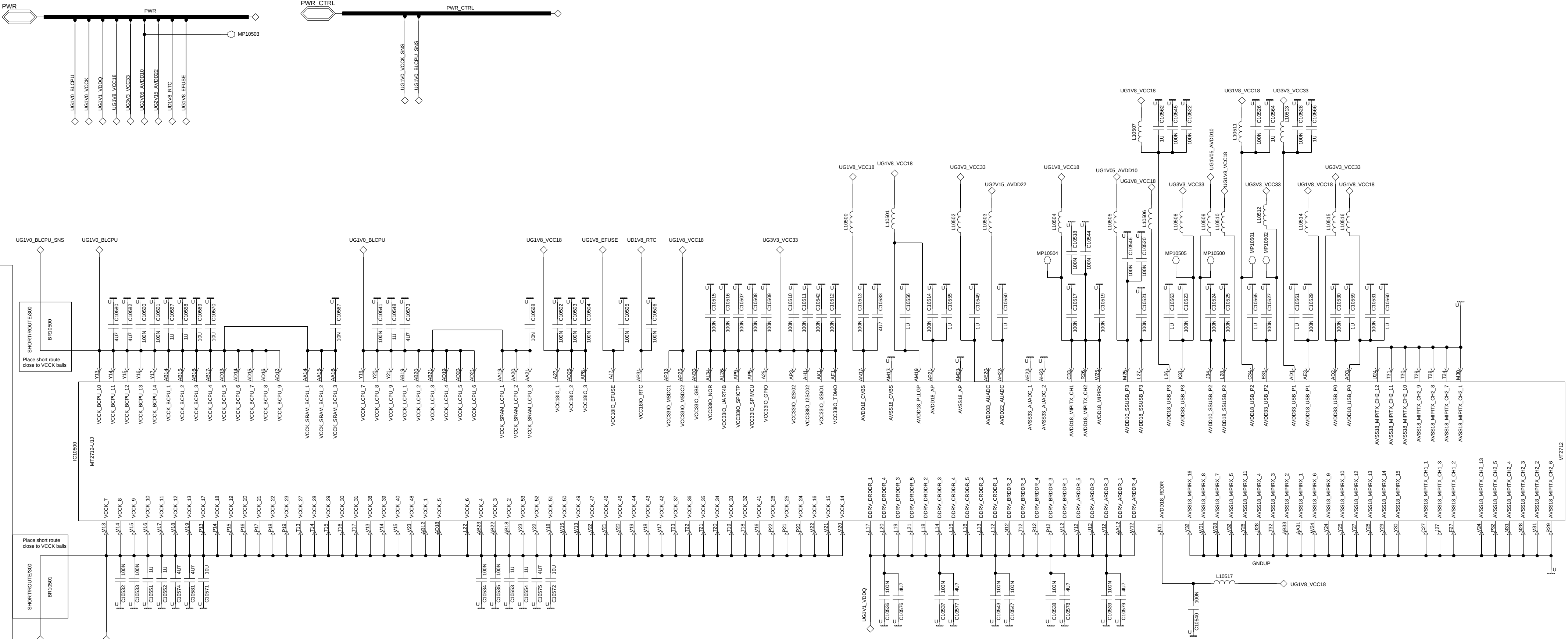
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schematic		Sheet 20	Sheets of 31	Sample	
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substitute for:		substitute by:			

SOC_LPDDR4 (10450-10499)



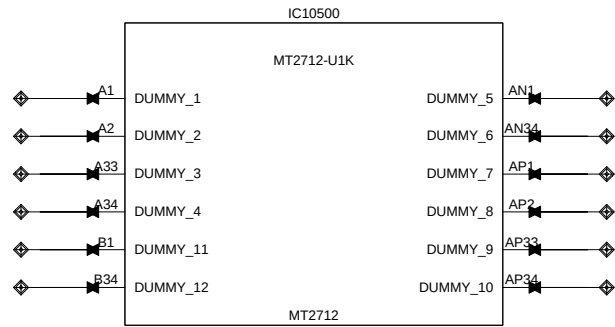
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4010_SoC_LPDDR4_chC+D				21	31	
substitute for:		substitute by:		Type	Variant	Part Number

SOC_POWER (10500-10599)



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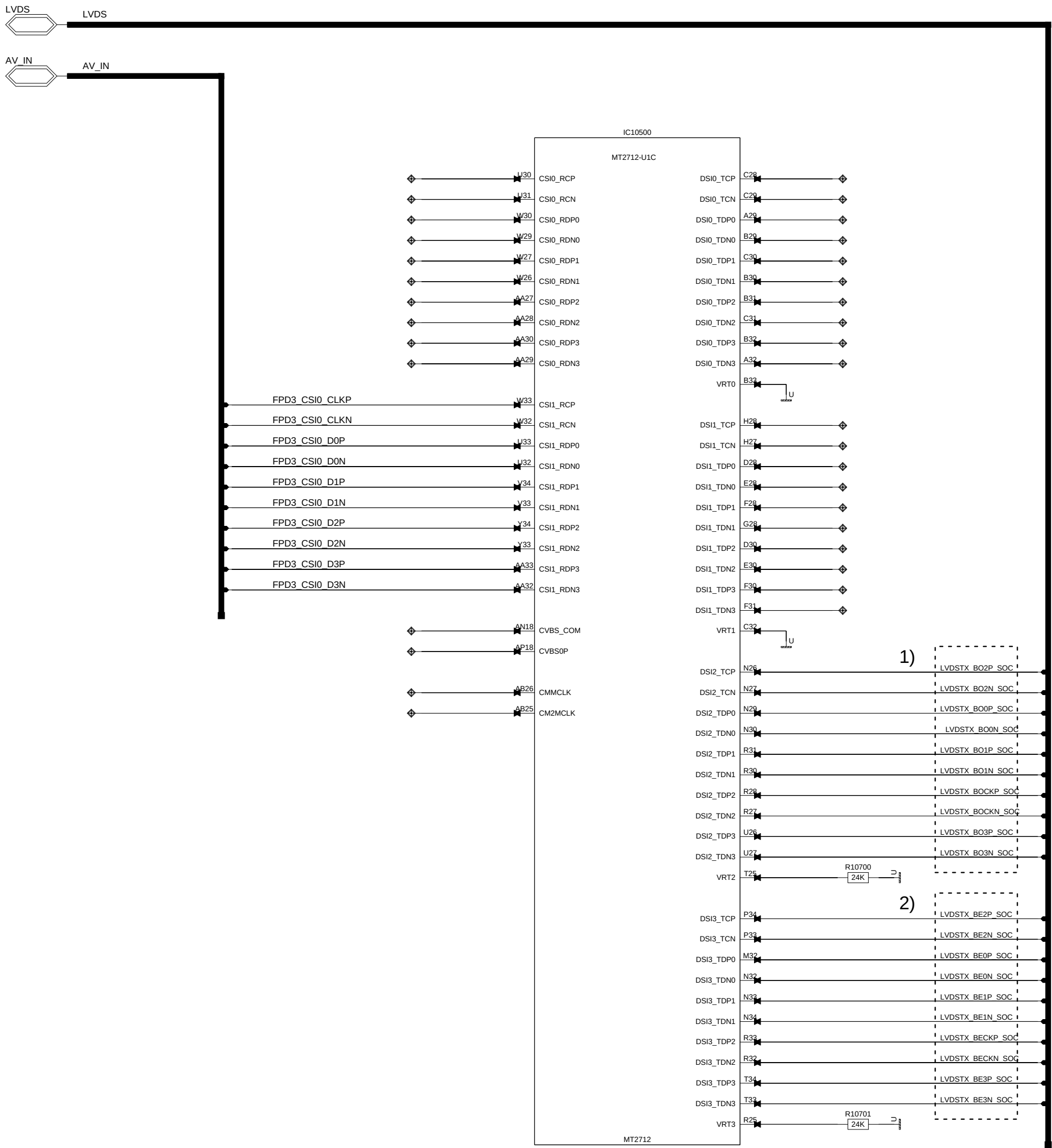
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			checked		
schematic			Sheet	Sheets of	Sample
4010_SoC_Power			22	31	
substitute for:			Type	Variant	Part Number
substitute by:					



Pin connection diagram for the MT2712-U1E IC. The diagram shows various external components connected to the IC pins. On the left, components include SOC_CTRL, IOC, BT, DEBUG, AV_IN, PWR_CTRL, and TOUCH. On the right, components include TP_SDA, TP_SCL, SOC_SDA2, SOC_SCL2, FPD3 I2C5 SDA, FPD3 I2C5 SCL, DEBUG SOC UART RXD, DEBUG SOC UART TXD, BT URXD3, BT UTXD3, BT URTS3, BT UCTS3, SOC_FPGA MISO EEPROM, SOC_FPGA MOSI EEPROM, SOC_FPGA SCK EEPROM, SOC_FPGA CSB EEPROM, IOC SPI4 MI, IOC SPI4 MO, IOC SPI4 CK, and IOC SPI4 CS. The IC pins are labeled with names like I2C_SDA0, I2C_SCL0, I2C_SDA1, I2C_SCL1, I2C_SDA2, I2C_SCL2, I2C_SDA3, I2C_SCL3, I2C_SDA4, I2C_SCL4, I2C_SDA5, I2C_SCL5, URXD0, UTXD0, URXD1, UTXD1, URXD2, UTXD2, URXD3, UTXD3, URTS3, UCTS3, URXD4, UTXD4, URTS4, UCTS4, URXD5, UTXD5, URTS5, UCTS5, SPI0_MI, SPI0_MO, SPI0_CK, SPI0_CSN, SPI1_SI, SPI1_SO, SPI1_CK, SPI1_CSN, SPI2_MI, SPI2_MO, SPI2_CK, SPI2_CSN, SPI3_MI, SPI3_MO, SPI3_CK, SPI3_CSN, SPI4_MI, SPI4_MO, SPI4_CK, SPI4_CSN, SPI5_MI, SPI5_MO, SPI5_CK, SPI5_CSN. The IC is labeled MT2712-U1E.

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schematic			Sheet 24	Sheets of 31	Sample
4010_SoC_Serial_IF			Type	Variant	Part Number
substitute for:		substitute by:			

SOC_VIDEO (10700-10799)

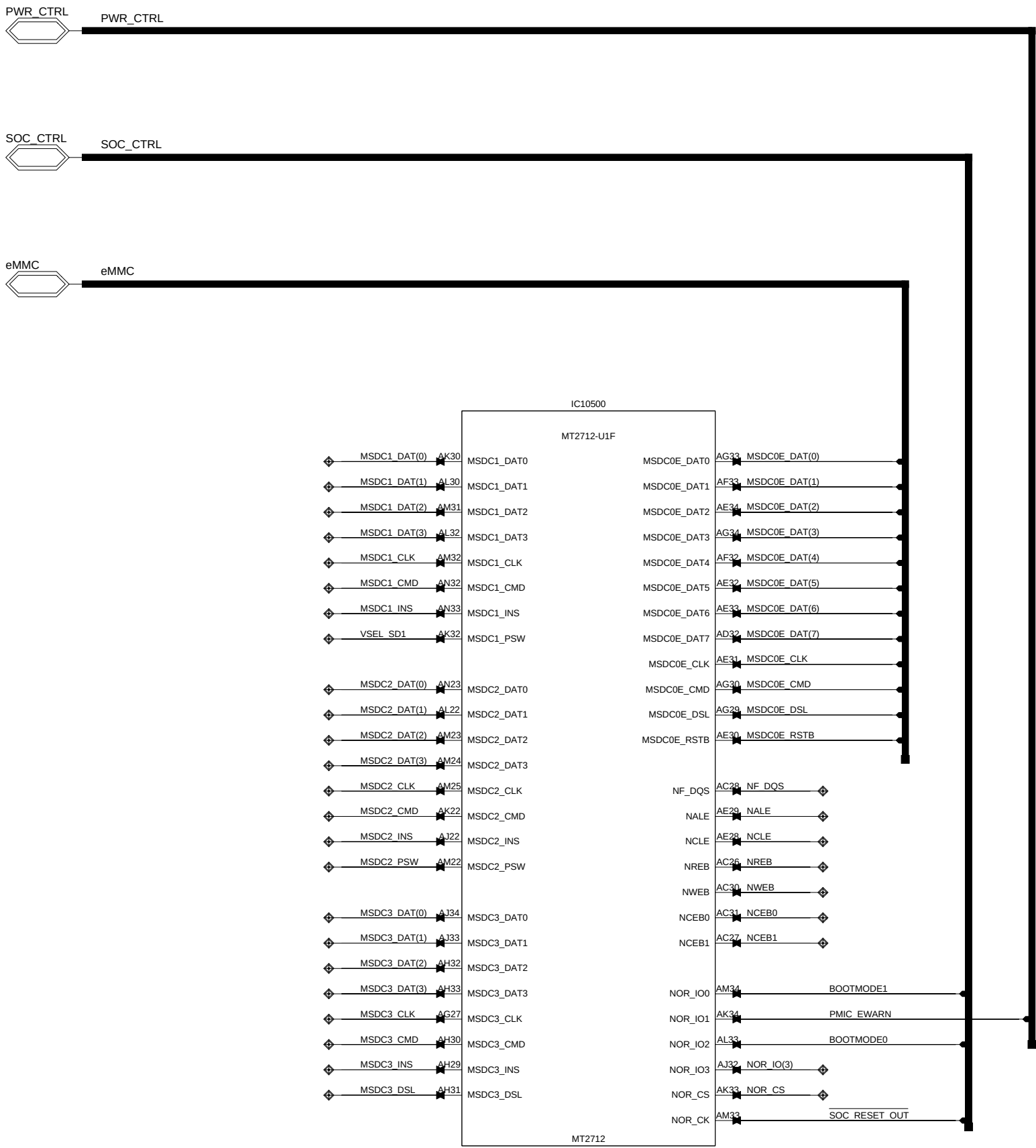


- 1) - Route as differential pair signals.
- Differential impedance: 100ohm (±10%)
- Maximum intra-pair skew: 0.127mm (5mils)
- Maximum skew from clock (BOCK) to data (BO[0-3]): 5mm

- 2) - Route as differential pair signals.
- Differential impedance: 100ohm (±10%)
- Maximum intra-pair skew: 0.127mm (5mils)
- Maximum skew from clock (BECK) to data (BE[0-3]): 5mm

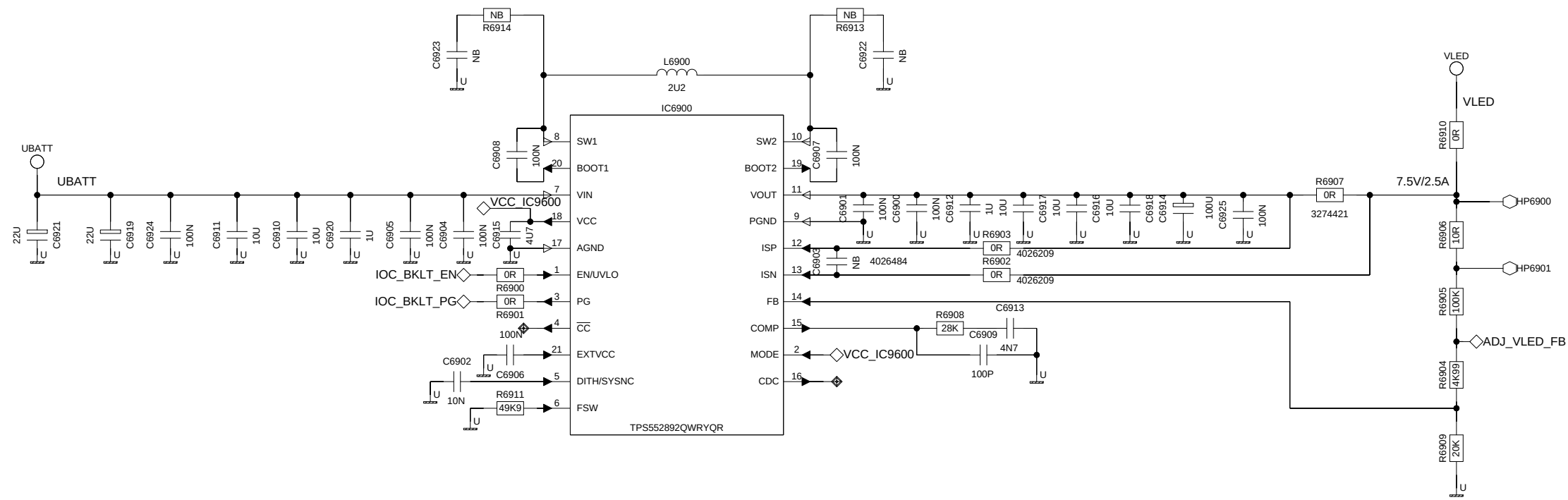
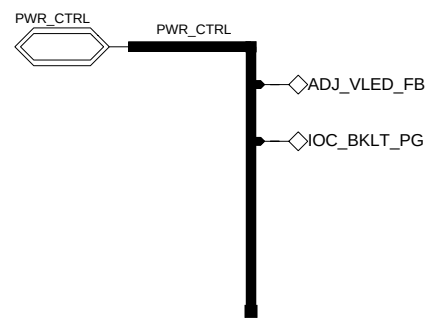
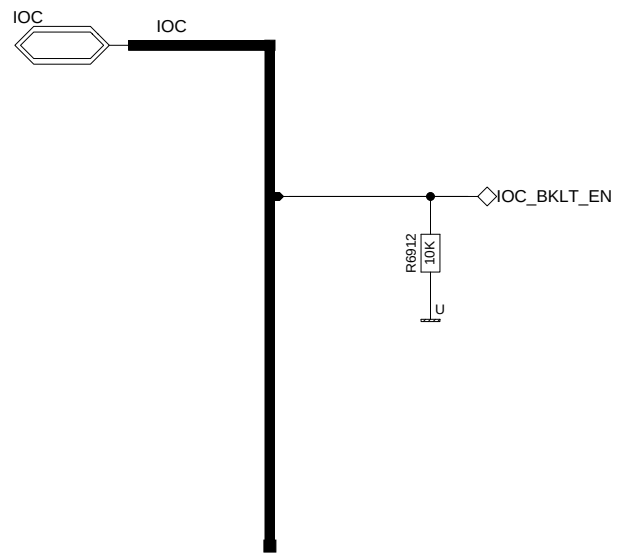
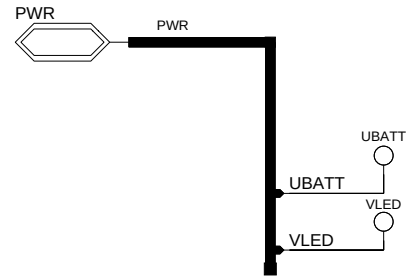
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						checked			
schematic			Sheet	Sheets of	Sample				
4010_SoC_Video			25	31					
substitute for:			Type	Variant	Part Number				
substitute by:									

SOC_eMMC_NAND_SID (10100-10199)



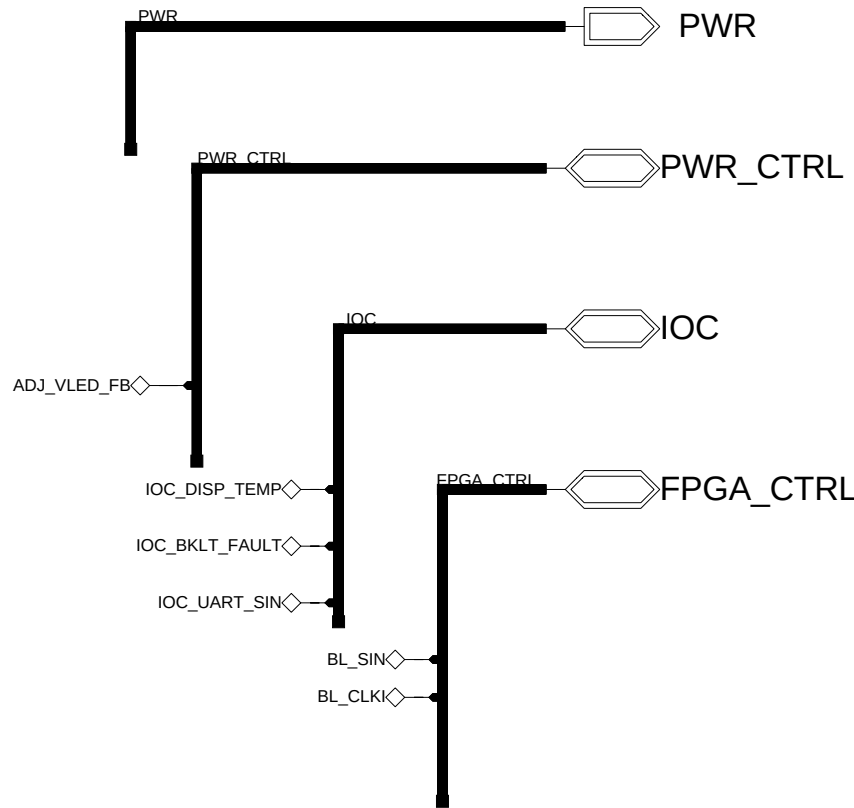
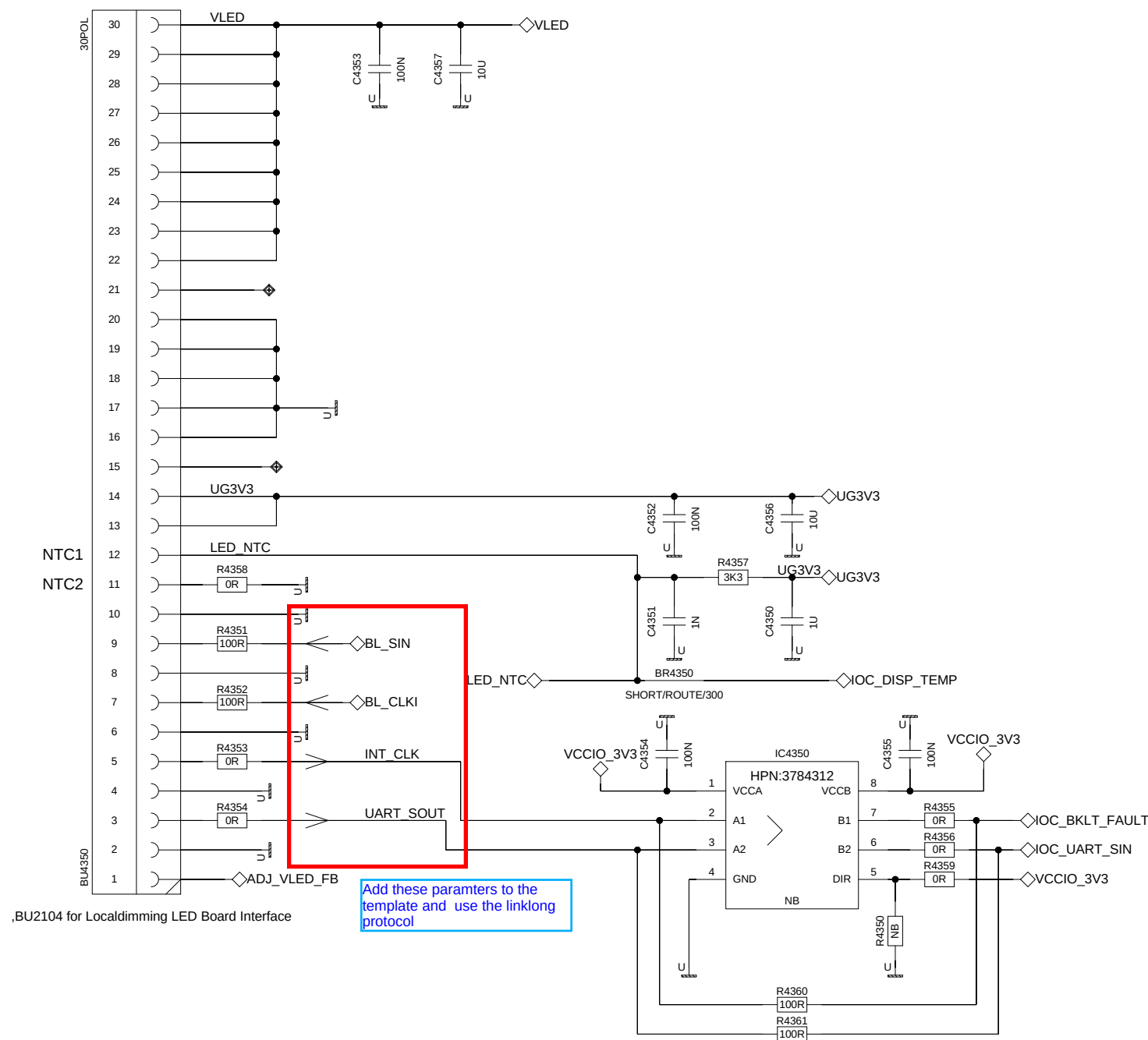
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schematic			Sheet	Sheets of	Sample	
4010_SoC_eMMC_NAND_SD			26	31		
substitute for:		substitute by:	Type	Variant	Part Number	

TFT BL (6900-6999)

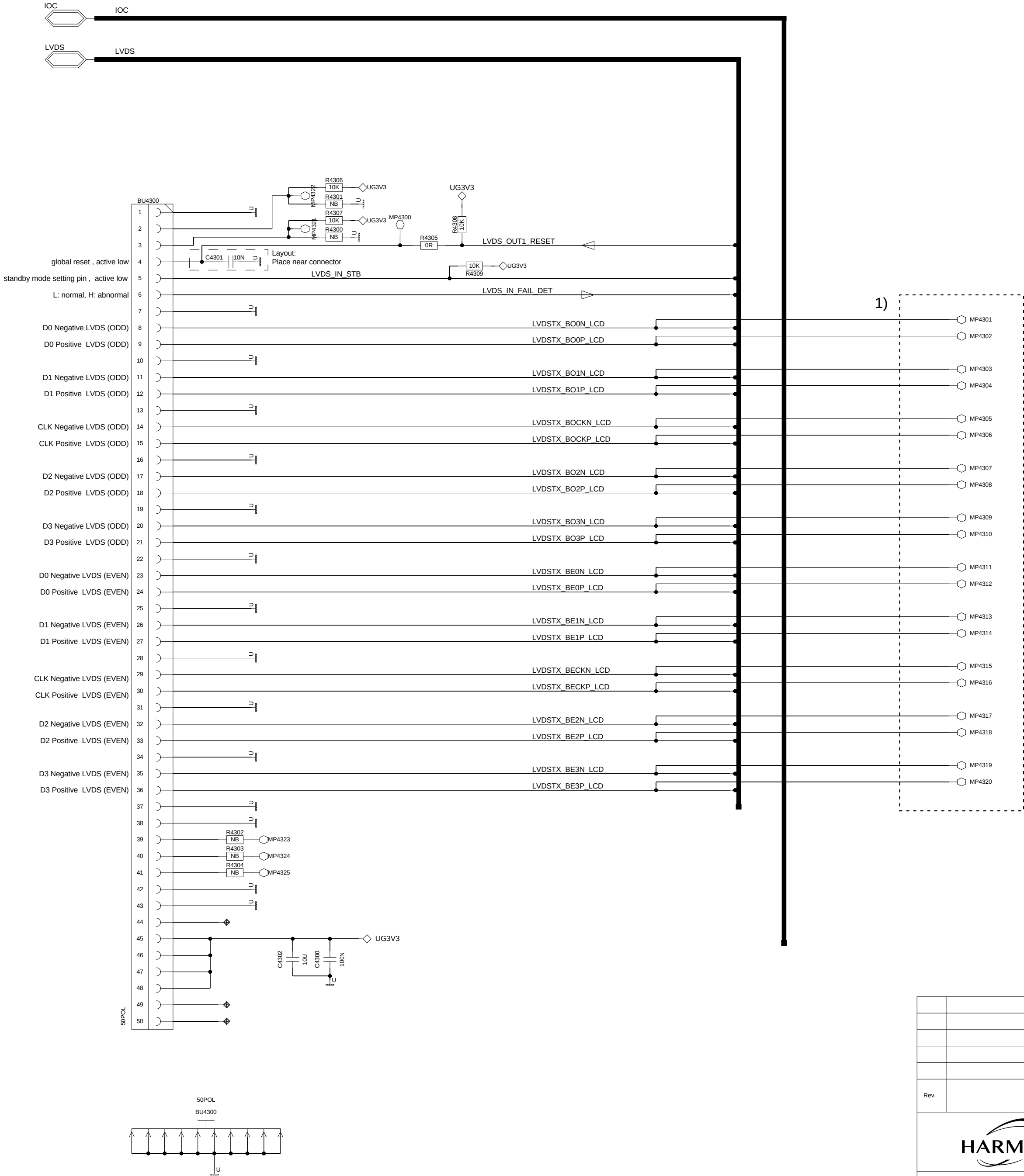
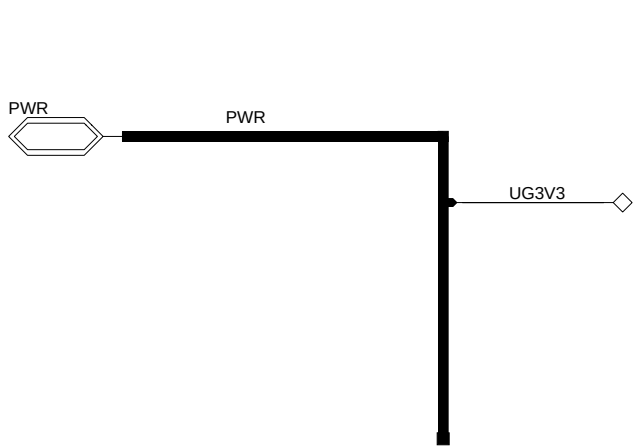


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				created					
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schematic					Sheet	Sheets of	Sample		
4010_TFT_BL					27	31			
					Type	Variant	Part Number		
substitute for:			substitute by:		C143				

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schematic		Sheet	Sheets of	Sample	
4010_TFT_Bl_Conn		28	31		
substitute for:		Type	Variant	Part Number	
			C143		

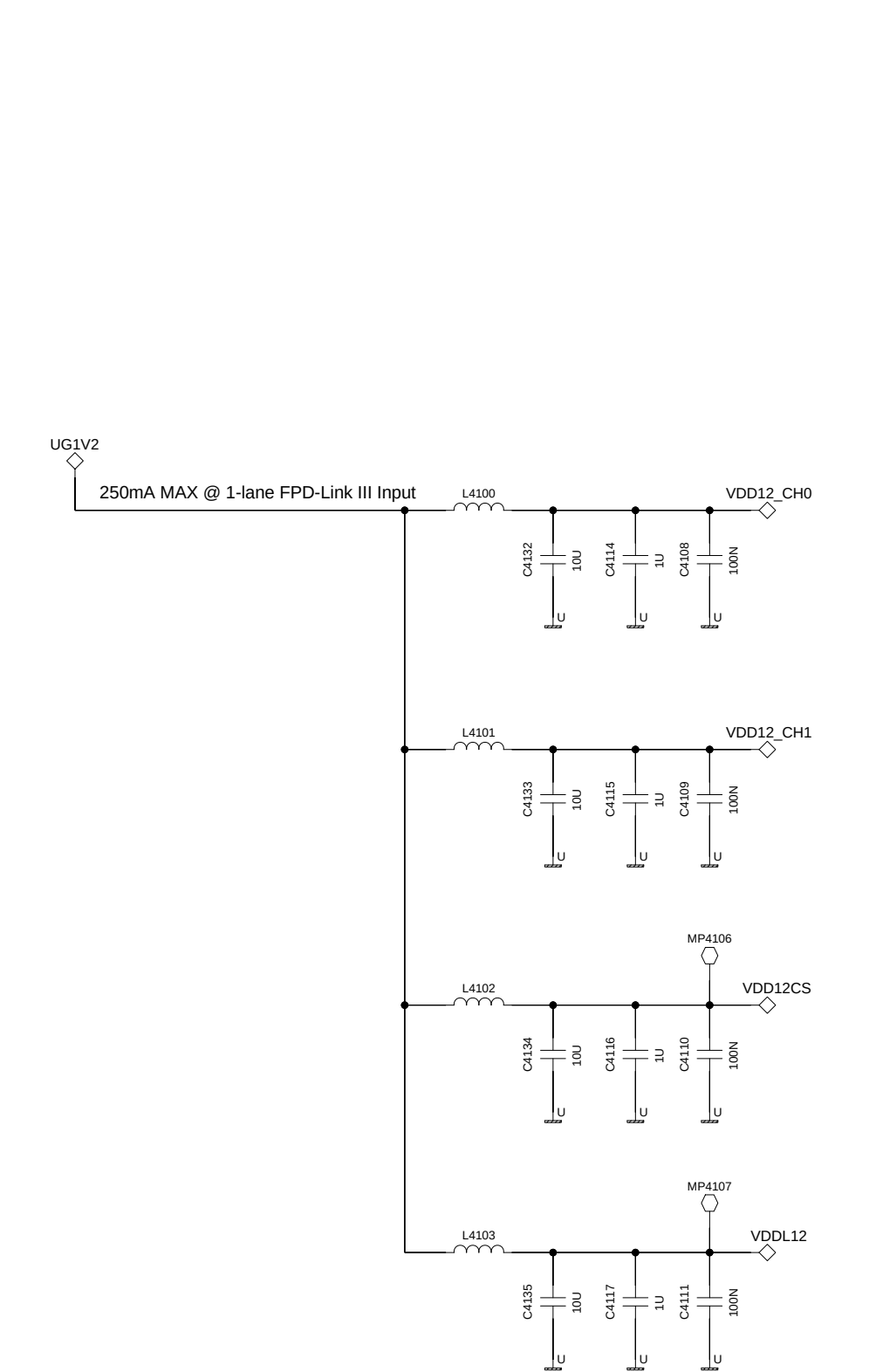


Verify pin out during layout

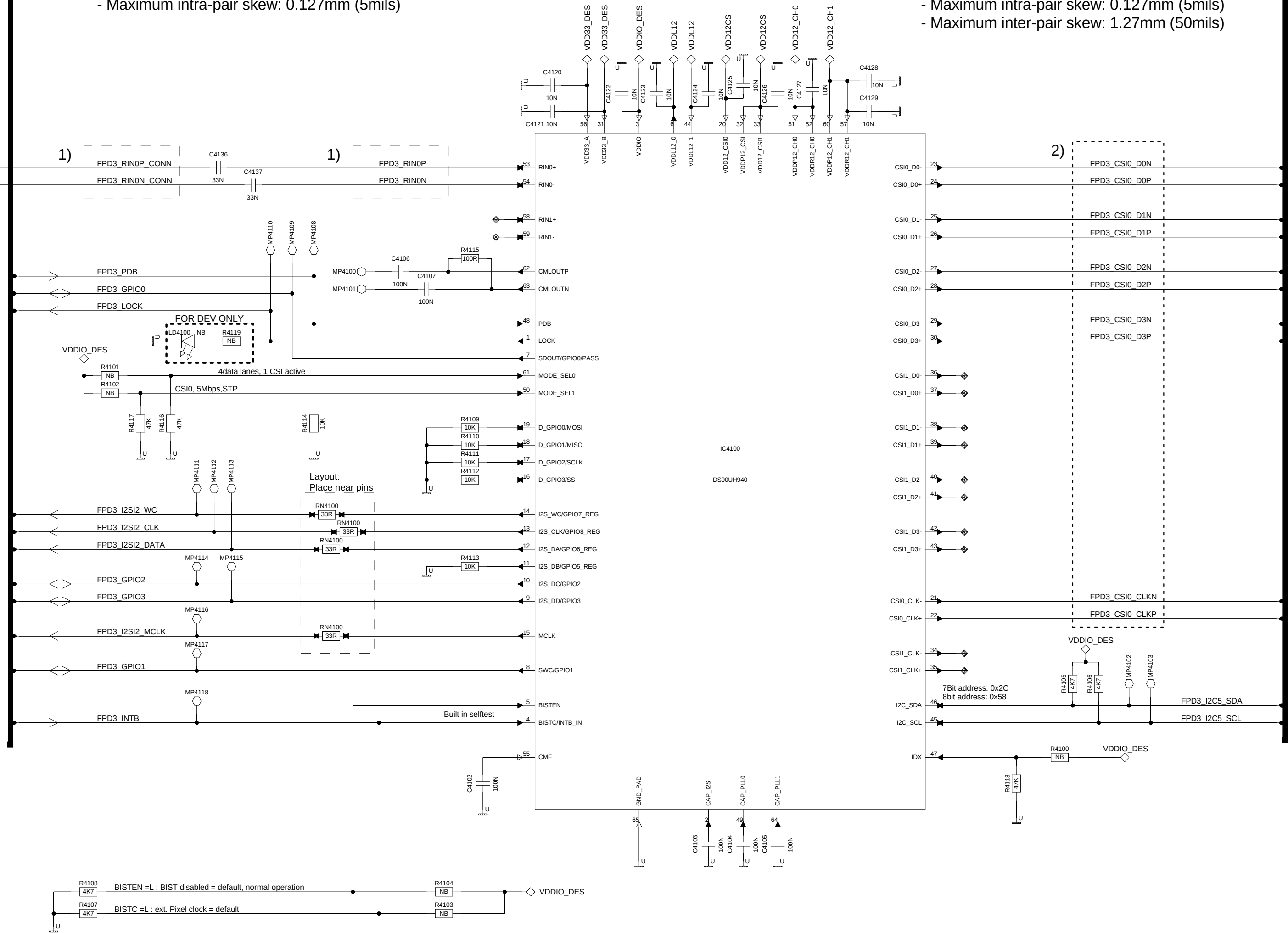
1) Testpoints should be placed as close as possible to an existing VIA

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				checked	
schematic			Sheet 29	Sheets of 31	Sample
4010_TFT_Conn			Type	Variant C143	Part Number
substitute for:		substitute by:			

The schematic diagram illustrates the power and signal connections for the AV_IN and PWR pins. The AV_IN section shows a 6mA MAX @ 1-lane FPD-Link III Input connected to VDDIO_DES, with decoupling capacitors C4130, C4112, C4100, C4118, and C4105. The PWR section shows a 122mA MAX @ 1-lane FPD-Link III Input connected to VDD33_DES, with decoupling capacitors C4131, C4113, C4101, and C4119. The diagram also shows the connection of the AV_IN and PWR pins to the MP4104 and MP4105 chips.

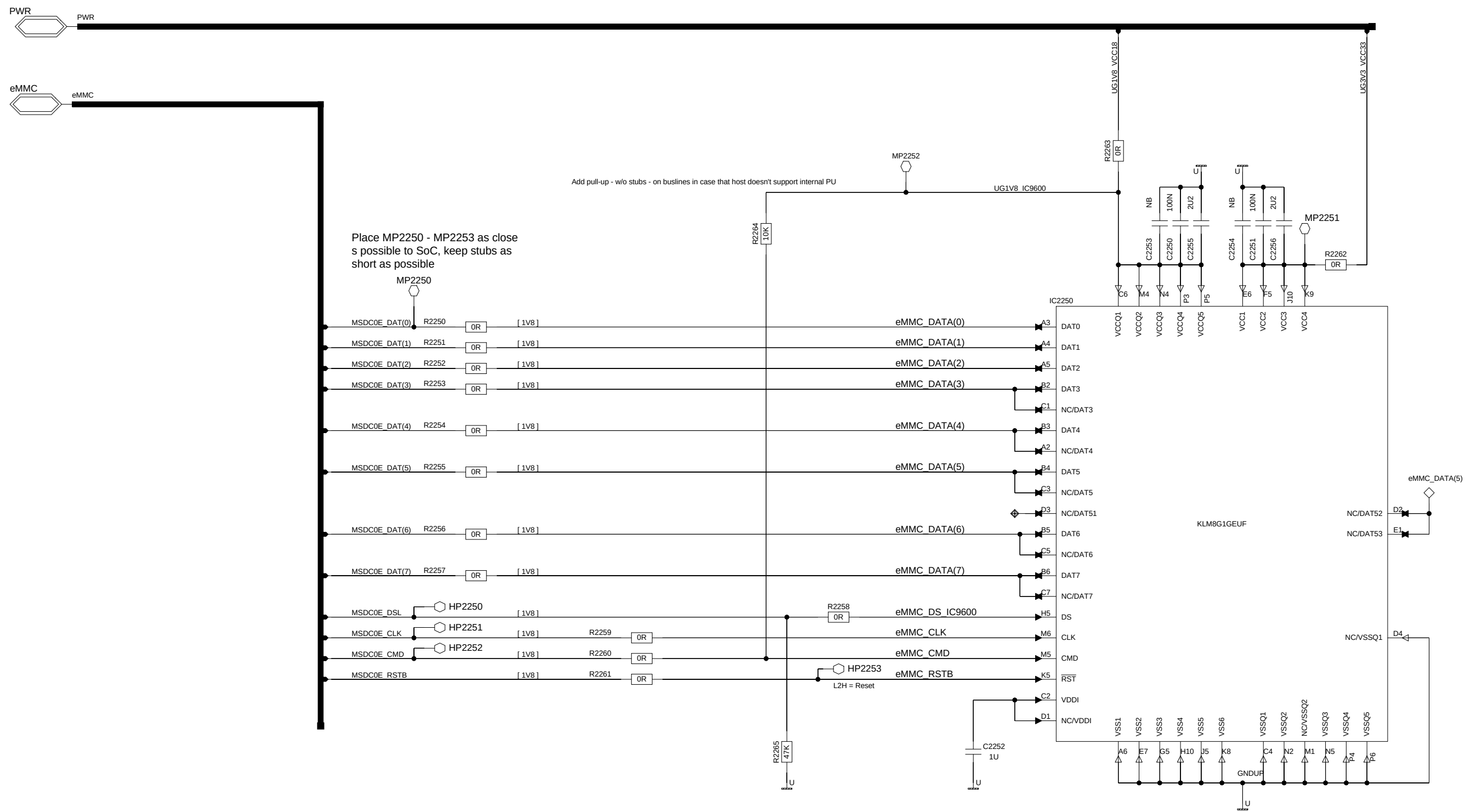


- 2) - Route as differential pair signals.
- Differential impedance: 100ohm ($\pm 10\%$)
- Maximum intra-pair skew: 0.127mm (5mils)
- Maximum inter-pair skew: 1.27mm (50mils)



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			checked		
schematic			Sheet	Sheets of	Sample
			30	31	
4010_VIDEO_DES			Type	Variant	Part Number
substitute for:		substitute by:		C143	

eMMC (2250-2299)



eMMC	Density	Part Number
Micron (-40 to 105°C)	8 GB	3881466
Micron (-40 to 85°C)	8 GB	3543617